Rate-Optimal Community Detection by Spectral Likelihood Decoding

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October 18, 2026

Abstract

The oracle likelihood-ratio test for a stochastic block model classifies a node using its edge counts to the true communities. We study how much of this evidence is retained by a spectral representation together with the observed degrees. We first give the simpler unregularized construction: for every fixed number of communities and $p = \Omega(\log n/n)$, the spectral likelihood decoder achieves the Zhou–Li benchmark $\mathbb{E}\,\mathrm{Mis} \le C\eta^*$ under comparable, uniformly well-conditioned block probabilities. Here η^* is the largest pairwise total variation affinity, equivalent under these assumptions to $e^{-I_n}/\sqrt{np}$, and $p = \max_{a,b} P_{ab}$. The proof controls likelihood tails after spectral projection, clipping and fitting, including the oracle polynomial factor. We then introduce the general algorithm retaining (D, U, Λ), where $D = A\mathbf{1}$ and the eigenpairs come from a degree-regularized adjacency matrix. Exact matrix identities allow its reconstructed block counts to sum to each observed degree. Initialization and refinement use one complete Bernoulli row criterion, calculated entirely from the retained information. A sharp transfer proposition gives the same $C\eta^*$ conclusion under explicit conditional tail hypotheses without imposing logarithmic expected degree; verifying these hypotheses for the full regularized algorithm remains open. Empirical reconstructed counts need not equal the original integers, and the comparison permits a constant-factor change in classification risk.

1 Introduction

The likelihood-ratio test gives a natural reference point for community detection in a stochastic block model (SBM). We use community detection as the broad task name, and recovery to describe the accuracy of the estimated labels. If the other labels and the block probabilities are known, classifying a node requires only its edge counts to the communities. Each count is weighted by the appropriate log-odds contrast, and the contribution of absent edges supplies the remaining term. The statistical question in this paper is how accurately this familiar test can be implemented from observed degrees and a small collection of spectral eigenpairs. We introduce the easier unregularized case first, where the eigenpairs alone suffice for the proved constant-factor oracle comparison. Section 4 then gives the general construction using both degrees and regularized spectral information, including the regime below logarithmic expected degree.

We treat the partition $z \in [K]^n$ as a fixed unknown parameter. The graph has independent upper-triangular Bernoulli edges with probabilities $P_{z_i z_j}$ and no self-loops [6]. Write $p = \max_{a,b} P_{ab}$. Our model uses the size bound $n_a \ge n/(\beta K)$, block probabilities comparable to p and bounded away from one, and a quantitative full-rank condition on P. For the unregularized theorem we additionally assume $p = \Omega(\log n/n)$. The general algorithm retains degrees and regularized eigenpairs and is defined also when $np \to \infty$ grows more slowly than $\log n$. Neither the normalized block matrix nor

the community proportions need converge. After compression, initialization and refinement use only the retained statistic: (U, Λ) in the first case and (D, U, Λ) in the general case.

The oracle test and the information retained by projection. Let Z be the membership matrix. The entries of AZ are the node-to-community edge counts. The oracle Bernoulli log-likelihood ratio between labels a and b is a linear combination of these counts plus a known nonedge term: its weight vector is constant on every community. When P has full rank, the signal space of $ZPZ^\top$ is exactly $\mathrm{col}(Z)$. If Π is the projector onto this population space, then

$$(A\Pi)Z = AZ.$$

Thus an oracle projection would preserve the counts exactly. Its real-valued entries are simply community averages, from which the integer sums can be recovered by multiplication. This observation explains why a subspace selected by a quadratic approximation criterion can still carry the evidence used by a likelihood-ratio test.

The algorithm uses an estimated subspace rather than Π. Its signed rank-K reconstruction is $\widetilde{A} = A\widehat{\Pi}$, where $\widehat{\Pi}$ is the projector calculated from the retained eigenvectors. Consequently its community sums differ from the observed counts by $A(\widehat{\Pi} - \Pi)Z$. These reconstructed sums are generally nonintegers; our analysis does not place each of them within $1/2$ of the corresponding count or recover the counts by rounding. The relevant criterion is instead how their errors affect classification. Relative count consistency alone does not control errors at the oracle risk scale. We instead control the likelihood tails at the scale required by the sharp bound of Zhou and Li.

The rate to be preserved. For the exact incident-edge Bernoulli test, let $I_{i,ab}$ be its Chernoff information and let I_n be the minimum over admissible nodes and distinct labels. Under our model bounds, $I_{i,ab} \asymp np$, the optimizing Chernoff parameter stays away from both endpoints, and the variance under the optimizing exponential tilt is of order np. The sharp Chernoff bounds of Zhou and Li [12] therefore give the pairwise oracle error scale

$$\frac{e^{-I_{i,ab}}}{\sqrt{np}}.$$

Our first theorem proves, for the complete unregularized algorithm,

$$\mathbb{E}\ \mathrm{Mis}(\widehat{z}, z) \le C\eta^*. \tag{1.1}$$

Here η^* is the maximum pairwise total variation affinity, defined in (3.3), and C depends only on the fixed model and algorithm bounds. Under our assumptions, $\eta^* \asymp e^{-I_n}/\sqrt{np}$. Throughout, rate optimality refers to this constant-factor comparison with the Zhou–Li oracle scale. It does not assert an error ratio converging to one, exact recovery of the original edge counts, or statistical sufficiency of the retained spectrum for all inference on z. We also do not prove a new global minimax lower bound.

The easier unregularized decoder. We retain the K eigenpairs of the original adjacency matrix with largest absolute eigenvalues, allowing signal eigenvalues of either sign. Clipping their reconstruction gives connection profiles q_i in a probability interval whose endpoints are determined by the retained eigenvalues. A complete Bernoulli row score compares these profiles to fitted connection profiles and includes the absent-edge term. This term matters for our density range:

Section 3 gives a dense example in which a sparse-score substitute loses a fixed fraction of the oracle exponent even with known parameters.

The same Bernoulli row score drives global-mean growing, component replacement and EM refinement. Growing adds an observed profile according to its total positive likelihood gain and jointly refits the current components. The algorithm then tests all deletion positions and their best observed replacement candidates for up to $\lceil \log n \rceil$ rounds, retaining the fit with the greatest common working mixture likelihood. This construction supplies a deterministic initialization guarantee without requiring a Euclidean initializer or a global optimum of an EM fit. The gain used to select a candidate and the soft mixture objective aggregate the common row score differently; their relation is part of the proof. One mandatory final E-step and profile/weight M-step precedes classification.

Why the spectral loss is small enough. Two parts of the argument connect this algorithm to the oracle test. First, a population replacement removes at least $1/K$ of the current hard fitting loss. A fitted profile unused by the true classes can be replaced without harming them; if every fitted profile is used, matching is a bijection and the profile assigned the largest loss can be replaced. Good observed candidates transfer this contraction to the empirical working objective. Its resulting small-loss certificate implies matched profile consistency and, more strongly, an E-step responsibility error of order $1/(np)$. The algorithm's existing final M-step therefore reaches the parameter accuracy required by the sharp likelihood analysis. No additional refinement step is needed for this objective-certified endpoint.

Second, we compare the spectral classification evidence to the oracle incident-edge evidence in the tail where errors occur. Conditioning on the graph with one vertex deleted gives an arrowhead expansion of the full-graph spectral reconstruction. It expresses each relevant spectral row sum as a linear function of the independent incident edges plus a bounded remainder. Replacing the population weights by the estimated projected weights changes the conditional log moment generating function by only a bounded amount at the oracle tilt. Clipping and parameter fitting require additional bounds; they cannot be discarded solely because the reconstruction is consistent. A finite collection of clipping branches and a local concentration bound under exponential tilting preserve the $(np)^{-1/2}$ factor. Exponential-confidence spectral events ensure that exceptional graphs are negligible even when np is much larger than $\log n$.

These comparisons allow a constant-factor change in the error probability. They do not require every spectral count to equal an observed integer or every classification decision to agree with the oracle. The substantive conclusion is that the remaining perturbation is small enough at the testing boundary to retain the full rate in (1.1).

Relation to existing recovery methods. Abbe and Sandon [1] characterize exact recovery in the general logarithmic-density SBM through Chernoff–Hellinger divergence and provide efficient algorithms reaching the threshold. Likelihood refinement after a spectral initializer is also well established. Amini, Chen, Bickel and Levina [3] fit mixtures to blockwise counts by pseudo-likelihood. Gao, Ma, Zhang and Zhou [5] obtain optimal misclassification performance from a weakly consistent initializer and penalized local likelihood refinement. Zhou and Amini [11] analyze spectral initialization and pseudo-likelihood classification in bipartite recovery, together with EM-type updates. A recent preprint of Wang [8] studies spectrally initialized variational EM in a general SBM and obtains a leading Chernoff exponent while estimating parameters and reusing the graph. Our retained information consists of degrees and spectral eigenpairs. The general refinement uses this compressed statistic, with no subsequent access to individual adjacency entries.

Spectral methods themselves attain optimal recovery guarantees in important settings. Abbe, Fan, Wang and Zhong [2] give entrywise eigenvector approximations and establish optimal thresholds and misclassification exponents for an untrimmed spectral method in the equal-sized symmetric two-community SBM at logarithmic density. For homogeneous planted-partition models with multiple possibly unequal communities, Zhang [9] obtains matching algorithm-specific exponents for trimming followed by eigenvalue-weighted spectral clustering and K-means; that exponent is distinguished from the information-theoretic exponent. Suwan et al. [7] use Gaussian-mixture limits of adjacency spectral embedding for empirical Bayes membership inference. These works motivate statistical decoding of spectral representations. Our focus is the sharp rare-event comparison with the Bernoulli oracle for general full-rank block matrices. The data-driven regularization and consistency analysis of Zhou and Amini [10] motivates the general construction in Section 4. Our unconditional sharp theorem is proved first under $p = \Omega(\log n/n)$; the later section states exactly which conditional likelihood-tail bounds extend the sharp comparison below that range.

Degrees and the general construction. Let $D = A\mathbf{1}$ and let W be the positive diagonal regularizer determined by D. If $WAWU = U\Lambda$, then

$$A(WU) = W^{-1}U\Lambda, \qquad A\mathbf{1} = D.$$

Thus (D, U, Λ) gives the exact action of A on the augmented space spanned by $\mathbf{1}$ and WU. Section 4 uses this action to construct degree-preserving row evidence and block counts. The constant-direction part of every likelihood contrast is then retained exactly. The same Bernoulli row criterion drives likelihood initialization and EM refinement. The general algorithm and its exact identities require no logarithmic density bound; a complete sparse sharp-rate theorem additionally requires the conditional tail hypotheses stated there.

Algorithmic scope and numerical evidence. The unregularized theorem covers the complete growing-and-replacement algorithm and every fixed K. General-K convergence of growing alone is not established. A five-community population example illustrates the possible loss of a class when growing uses only one EM update per stage. The numerical section reports a 60-graph benchmark of the complete Bernoulli implementation and, separately, the earlier 240-graph sparse-score experiments. The complete benchmark shows no additional classification improvement from replacement in that collection and exposes finite-sample difficulties at weak signals. It complements the theorem without establishing oracle risk ratios or uniform empirical superiority over K-means.

Section 2 introduces the simpler unregularized algorithm and its relation to the oracle test. Section 3 proves its sharp guarantee. Section 4 introduces the general degree-plus-spectrum algorithm and its sharp transfer criterion. Section 5 reports the experiments, and Section 6 concludes. Proofs and implementation details are in the appendix.

2 The unregularized case: from oracle likelihood ratios to spectral decoding

2.1 Model and retained information

Let A be the adjacency matrix of an undirected graph without self-loops. For a fixed unknown partition $z \in [K]^n$, the upper-triangular entries are independent, with

$$A_{ij} \sim \text{Bernoulli}(P_{z_i z_j}), \quad i < j, \qquad A_{ji} = A_{ij}, \quad A_{ii} = 0. \tag{2.1}$$

We follow Zhou and Li [12]: P is the symmetric block probability matrix, $p_{aj} = P_{a,z_j}$, and p_{a*} is the connection profile of community a. Write $Z_{ia} = \mathbf{1}\{z_i = a\}$, $n_a = \sum_i Z_{ia}$, $\pi_{a,n} = n_a/n$, and

$$p = \max_{a,b} P_{ab}.$$

The known community number $K \geq 2$ is any fixed integer. For this unregularized case, the model and density conditions are, for a fixed constant $\beta \geq 1$ and fixed positive constants κ, η,

$$n_a \geq \frac{n}{\beta K}, \qquad P_{ab} \geq \kappa p \quad (a, b \in [K]), \qquad \sigma_{\min}(P) \geq \kappa p, \tag{2.2}$$

$$p \leq 1 - \eta, \qquad p = \Omega(\log n/n). \tag{2.3}$$

We use the conventional imbalance parameter β, with the normalization $n/(\beta K)$ common in SBM recovery results [5, 12]. Only this lower size bound is required; a separate upper balance bound is unnecessary. Writing n/β instead would absorb the fixed factor K into β. These conditions are used for the unregularized sharp theorem. Section 4 treats the general retained statistic and replaces the logarithmic density condition by $np \to \infty$ in its sparse-rate target. We may take $\kappa \leq 1$. Neither P/p nor the community proportions need converge. The probability matrix may vary with n within these bounds. In particular, there is no requirement that $(n/\log n)P$ be fixed.

For $z_i = a$, the expected degree is

$$\mathbb{E}[\deg(i) \mid z] = \sum_b (n_b - \mathbf{1}\{a = b\}) P_{ab}. \tag{2.4}$$

These expectations lie between $\kappa(n-1)p$ and np and may differ across communities. Thus they are uniformly of order np, and (2.3) implies that all expected degrees are $\Omega(\log n)$. The density condition permits intermediate densities and constant edge probabilities. The probability-comparability and singular-value conditions ensure a uniformly separated, well-conditioned signal; degree alone does not imply identifiability.

Retain the K eigenpairs of the original adjacency matrix with largest absolute eigenvalues:

$$AU = U\Lambda, \qquad U^\top U = I_K, \qquad \widetilde{A} = U\Lambda U^\top. \tag{2.5}$$

Both signs of signal eigenvalues are allowed. In this unregularized case, every subsequent operation uses only (U, Λ), including initialization, fitting, replacement and classification. The degree vector is retained explicitly for the general construction in Section 4.

2.2 The oracle LRT depends on community edge counts

For testing $z_i = a$ against $z_i = b$ with the other labels and P known, put

$$w_j^{ab} = \log \frac{p_{aj}(1 - p_{bj})}{p_{bj}(1 - p_{aj})}, \qquad c_i^{ab} = \sum_{j \neq i} \log \frac{1 - p_{aj}}{1 - p_{bj}}. \tag{2.6}$$

The exact incident-edge Bernoulli log-likelihood ratio is

$$L_i^{ab} = A_i w^{ab} + c_i^{ab} = \sum_{c=1}^{K} b_{ic} w_c^{ab} + c_i^{ab}, \qquad b_{ic} = \sum_{j : z_j = c} A_{ij}, \tag{2.7}$$

where $w_c^{ab} = \log\{P_{ac}(1 - P_{bc})/[P_{bc}(1 - P_{ac})]\}$ is the common value of w_j^{ab} in community c. Thus the oracle LRT needs K community edge counts for a node, rather than the individual locations of

its edges within each community. Following Amini, Chen, Bickel and Levina [3, Section 2.1], for a candidate labeling e write $b_{ic}(e) = \sum_j A_{ij}\mathbf{1}\{e_j = c\}$; here $b_{ic} = b_{ic}(z)$. The matching spectral count is $\widetilde{b}_{ic}(e) = \sum_j \widetilde{A}_{ij}\mathbf{1}\{e_j = c\}$. These are node-to-block counts. Total ordered counts between two blocks are entries of $Z(e)^\top AZ(e)$; an entry on its diagonal counts each undirected edge twice. The self-loop coordinate contributes no edge because $A_{ii} = 0$. Knowledge of the other labels is an oracle device for defining the benchmark; the decoder below is not supplied those labels.

2.3 What spectral compression retains and what it can lose

The population membership projector and its spectral estimate are

$$\Pi = Z(Z^\top Z)^{-1}Z^\top, \qquad \widehat{\Pi} = UU^\top.$$

Under the full-rank condition on P, the signal space $\mathrm{col}(ZPZ^\top)$ equals $\mathrm{col}(Z)$. It therefore contains every community indicator and every oracle weight vector w^{ab}. This is the algebraic reason for using a spectral representation to approximate the LRT.

Proposition 2.1 (Oracle projection and empirical spectral compression)**.** *For the oracle projector,* $\Pi Z = Z$ *and*

$$(A\Pi)Z = AZ, \qquad Z^\top(A\Pi)Z = Z^\top AZ. \tag{2.8}$$

For the retained empirical eigenpairs,

$$\widetilde{A} = A\widehat{\Pi}, \qquad \widetilde{A}_i w - A_i w = A_i(\widehat{\Pi} - \mathrm{Id})w \tag{2.9}$$

for every $w \in \mathbb{R}^n$*. In particular,*

$$\widetilde{b}_{ic} := \sum_{j:z_j=c} \widetilde{A}_{ij}, \qquad \widetilde{b}_{ic} - b_{ic} = A_i(\widehat{\Pi} - \Pi)\mathbf{1}_{\{z=c\}}. \tag{2.10}$$

Proof. The oracle identities follow from $\Pi Z = Z$. With $N = Z^\top Z$, the factorization

$$ZPZ^\top = (ZN^{-1/2})(N^{1/2}PN^{1/2})(ZN^{-1/2})^\top$$

identifies the signal space. Finally, $AU = U\Lambda$ gives $A\widehat{\Pi} = U\Lambda U^\top$ and the remaining identities. □

The distinction between the two projectors is statistical. For j in community c, the real-valued oracle entry is $(A\Pi)_{ij} = b_{ic}/n_c$: it stores the integer count as a block average, including its random fluctuation. The empirical quantity $\widetilde{b}_{ic}$ need not be an integer, and our result does not assert that its error is below $1/2$ or that rounding recovers b_{ic}. Small subspace error alone would not imply either assertion.

Instead, we control the loss relevant to the LRT. Substituting $\widetilde{b}_{ic}$ for b_{ic} perturbs a weighted likelihood contrast. Clipping and fitted parameters introduce further changes. The proof controls these changes through conditional exponential moments and concentration under the oracle Chernoff tilt, where classification errors occur. This control is strong enough to preserve the oracle error rate up to a constant factor, including its $(np)^{-1/2}$ factor. It does not require exact recovery of each count. Ordinary consistency of the counts relative to their means, or a score approximation holding only for typical graphs, would not suffice for this rare-event comparison.

The actual mean is

$$\mathbb{E}[A \mid z] = ZPZ^\top - \mathrm{diag}(P_{z_i z_i}). \tag{2.11}$$

The analysis includes this absent-diagonal correction and estimates the empirical signal space from the same graph used for classification. The oracle counts and projectors above are comparison tools, not additional inputs to the algorithm.

2.4 Observed profiles and the complete Bernoulli score

To define logarithms of reconstructed probabilities, set

$$s = \max\{1, \|\Lambda\|_{\mathrm{op}}\}, \qquad \varepsilon_n = c_0 s/n, \qquad q_{ij} = \mathrm{clip}(\widetilde{A}_{ij}, \varepsilon_n, 1-\varepsilon_n), \tag{2.12}$$

where $0 < c_0 < \min\{\kappa, \eta, 1\}/8$ is a fixed algorithm parameter and $\mathrm{clip}(x, l, u) = \min\{u, \max\{l, x\}\}$. The admissibility of c_0 is an explicit model condition on this chosen logarithm floor. The scale s and the floor are calculated from the retained eigenvalues. The reconstructed diagonal is retained. Since $s \le n$ for an adjacency matrix and $c_0 < 1/8$, the interval is nonempty even on exceptional graphs.

For $\widehat{p} \in (0,1)^n$, the working score is

$$\ell_i(\widehat{p}) = \sum_{j=1}^{n} \{q_{ij} \log \widehat{p}_j + (1 - q_{ij}) \log(1 - \widehat{p}_j)\}. \tag{2.13}$$

The second term retains the information in absent edges at every density. The q_i are dependent fractional profiles, so this is a working fitting criterion rather than the exact likelihood of independent edge observations. Its likelihood contrasts approximate the oracle contrasts at the error-probability scale established in Section 3. For fitted profiles $\widehat{p}_1, \dots, \widehat{p}_K$ and weights $\widehat{\pi} \in \Delta_K$, fit

$$\mathcal{L}(\widehat{\pi}, \widehat{p}) = \sum_i \log \sum_a \widehat{\pi}_a e^{\ell_i(\widehat{p}_a)}, \qquad \widehat{z}_i = \arg\max_a \{\ell_i(\widehat{p}_a) + \log \widehat{\pi}_a\}. \tag{2.14}$$

The output error is the permutation-invariant fraction $\mathrm{Mis}(\widehat{z}, z) = \min_{\sigma \in \mathfrak{S}_K} n^{-1} \sum_i \mathbf{1}\{\sigma(\widehat{z}_i) \neq z_i\}$.

2.5 Growing initialization and exact updates

Start one profile at the global mean $n^{-1} \sum_i q_i$. At stage $t = 2, \dots, K$, if the existing profiles are $\widehat{p}_1, \dots, \widehat{p}_{t-1}$, compute

$$h_i = \max_{a<t} \ell_i(\widehat{p}_a), \qquad \Delta(c) = \sum_i (\ell_i(q_c) - h_i)_+. \tag{2.15}$$

Append a node profile maximizing this all-node gain, excluding previously added node IDs, initialize the t weights uniformly, and jointly fit all t profiles. No observations are removed. The fitted profiles supply the next stage. Initialization and refinement use the same complete nodewise score (2.13). The positive hard gain selects candidate profiles, while the soft mixture objective (2.14) compares fitted models; these are different aggregations of the same nodewise score.

At any component count, the exact E/M updates are

$$R_{ia} = \frac{\widehat{\pi}_a e^{\ell_i(\widehat{p}_a)}}{\sum_b \widehat{\pi}_b e^{\ell_i(\widehat{p}_b)}}, \qquad N_a = \sum_i R_{ia}, \tag{2.16}$$

$$\widehat{p}_{aj}^{\,\mathrm{new}} = \frac{\sum_i R_{ia} q_{ij}}{N_a}, \qquad \widehat{\pi}_a^{\,\mathrm{new}} = N_a/n. \tag{2.17}$$

A component of zero responsibility mass retains its previous profile. Each fit performs at least one M-step. Further exact updates may use any prescribed finite iteration bound and any stopping rule within that bound. All means remain convex combinations of the same q rows, and each E/M pair increases the same working objective. The update formula is unchanged by including the nonedge term: maximizing a weighted Bernoulli cross-entropy still gives the weighted mean.

2.6 Repeated likelihood replacement for every fixed K

Repeated component replacement supplies a fitting certificate for every fixed K. Set $T_n = \lceil \log n \rceil$. In each round, refer all trials to the same current fit. For each component $r \in [K]$, delete its profile and compute

$$h_i^{(-r)} = \max_{a \neq r} \ell_i(\widehat{p}_a), \qquad c_r \in \arg\max_{c \in [n]} \sum_i (\ell_i(q_c) - h_i^{(-r)})_+. \tag{2.18}$$

Replace profile r by q_{c_r}, initialize the trial's weights uniformly, and run a finite exact EM fit. The candidate scan here includes every node ID, including previously used IDs. Compare the K trial endpoints and the unchanged current fit using $\mathcal{L}$, and keep the largest value. Ties are deterministic, with the current fit retained on an objective tie. The algorithm may stop after a full round with no improving trial; the contraction below certifies sufficiently small loss at that point.

After the replacement rounds, compute the selected fit's responsibilities and perform one mandatory profile and weight M-step before decoding. The objective certificate makes this E-step's total responsibility error $O(n/(np))$ after a common label matching. The following M-step then makes fitted likelihood contrasts accurate enough for the constant-factor oracle rate. Additional exact EM updates after this step are allowed.

Algorithm 1: Spectral likelihood with repeated parameter replacement

1. Form q from the retained eigenpairs using (2.12).
2. Obtain K fitted profiles by global-mean growing and joint EM.
3. For $T_n = \lceil \log n \rceil$ rounds, test all K deletion positions; choose each replacement node by (2.18), jointly fit it from uniform weights, and retain the greatest working likelihood across the trials and the old fit.
4. Perform one final E-step and profile/weight M-step, then classify by (2.14).

The replacement certificate controls all K fitted profiles simultaneously, including unequal communities and signal eigenvalues of either sign. It supplies the initialization needed for the sharp likelihood-tail comparison without assuming that growing alone reaches the correct basin. The general-K theorem concerns Algorithm 1 with its stated replacement budget and final E/M update.

2.7 Computational cost

Forming and storing q requires $O(n^2)$ storage. The complete node-candidate score matrix can be cached using $O(n^3)$ arithmetic and $O(n^2)$ storage. With current component scores cached, all K deletion scans in a round cost $O(Kn^2 + K^2 n)$: obtain the minimum and second minimum divergences of each row to handle every deletion. If each trial uses at most J EM updates, its fitting cost is $O(JKn^2)$. The $O(\log n)$ rounds therefore cost $O(JK^2 n^2 \log n)$ in addition to the candidate cache and growing fits. The direct implementation is polynomial in n for fixed K and polynomial J. It does not yet provide a subquadratic storage claim.

3 Sharp guarantees for the unregularized case

We compare the spectral decoder with the familiar oracle likelihood-ratio test. The issue is how much the empirical spectral compression changes the testing error, rather than whether its real-valued block sums recover the original integer counts. Throughout this section, the fixed parameter z and the block matrix P satisfy (2.2)–(2.3), and c_0 satisfies the condition following (2.12).

3.1 The oracle benchmark and the main guarantee

For $a \neq b$, the oracle reveals P and all labels other than that of vertex i, and tests the two product-Bernoulli laws with connection profiles $p_{a,-i}$ and $p_{b,-i}$. Their exact Chernoff information is

$$D_\alpha(p_{a,-i} \| p_{b,-i}) = -\sum_{j \neq i} \log\{p_{aj}^{1-\alpha} p_{bj}^{\alpha} + (1-p_{aj})^{1-\alpha}(1-p_{bj})^{\alpha}\}, \tag{3.1}$$

$$D_{i,ab}^* = \max_{\alpha \in [0,1]} D_\alpha(p_{a,-i} \| p_{b,-i}), \qquad I_n = \min_{i,\ a \neq b} D_{i,ab}^*. \tag{3.2}$$

The absent self-loop is excluded. Under our assumptions, $D_{i,ab}^* \asymp np$, its maximizing parameter is bounded away from zero and one, and the likelihood ratio has tilted variance of order np. Following Zhou and Li [12], let $\varphi_{i,a}$ be the product-Bernoulli mass function with profile $p_{a,-i}$, and define its total variation affinity with the alternative by

$$\eta_{i,ab} = \sum_{x \in \{0,1\}^{n-1}} \min\{\varphi_{i,a}(x), \varphi_{i,b}(x)\}, \qquad \eta^* = \max_{i, a \neq b} \eta_{i,ab}. \tag{3.3}$$

The equal-prior binary oracle risk is exactly $R_{i,ab}^{\mathrm{or}} = \eta_{i,ab}/2$. Their sharp Chernoff bounds give

$$\eta_{i,ab} \asymp \frac{e^{-D_{i,ab}^*}}{\sqrt{np}}, \qquad \eta^* \asymp \frac{e^{-I_n}}{\sqrt{np}}. \tag{3.4}$$

All comparisons are uniform over the stated model bounds. Fixed binary priors bounded away from zero and one change only their constants. Appendix A.1 verifies the uniform conditions used in this oracle comparison.

Theorem 3.1 (Spectral likelihood decoding preserves the oracle rate)**.** *For every fixed $K \geq 2$, Algorithm 1 satisfies*

$$\mathbb{E}\,\mathrm{Mis}(\widehat{z}, z) \leq C\eta^*. \tag{3.5}$$

The constant depends only on the fixed model bounds, the density lower-bound constant, K, and the admissible c_0. Initialization, fitting and classification use only the retained eigenpairs, and no weak-initialization assumption is imposed. Growing and replacement trials may use any finite number of exact EM updates, with at least one M-step per fit. The mandatory final E/M update is sufficient. Any additional finite sequence of exact E/M updates, including a finite data-dependent stopping rule, preserves the bound. Polynomial iteration caps give a polynomial-time algorithm.

In particular,

$$\mathbb{P}\{\mathrm{Mis}(\widehat{z}, z) > 0\} \leq Cn\eta^*,$$

so exact recovery holds whenever $n\eta^* \to 0$.

The theorem permits $p \to 0$, intermediate densities, and constant edge probabilities below one. It does not require convergence of P/p or of the community proportions. The comparison in (3.5) concerns the order of the optimal binary testing error. It allows a constant-factor change in risk, and proves neither recovery of every integer edge count nor a risk ratio tending to one. We also make no new global minimax lower-bound claim for an unspecified parameter class.

The sharp oracle affinity. More generally, the sharp Chernoff comparison of Zhou and Li has the form

$$\eta_{i,ab} \asymp \frac{e^{-D^*_{i,ab}}}{\alpha^*_{i,ab}(1-\alpha^*_{i,ab})\sqrt{V_{i,ab}}}, \tag{3.6}$$

where $\alpha^*_{i,ab}$ maximizes the Chernoff divergence and $V_{i,ab}$ is the variance of the log likelihood ratio under the optimizing exponential tilt. This comparison requires the common-support, third-moment and effective-variance conditions in their sharp bound. Our assumptions ensure these conditions uniformly, with $\alpha^*_{i,ab}$ bounded away from zero and one and $V_{i,ab} \asymp np$. Formula (3.6) describes the more general oracle benchmark; extending the spectral guarantee beyond our model conditions requires a separate perturbation analysis.

Corollary 3.2 (Planted-partition consequences of the oracle bound)**.** *Suppose, in addition to the assumptions of Theorem 3.1, that $P_{aa} = p_{\text{in}}$ and $P_{ab} = p_{\text{out}}$ for $a \neq b$. Let*

$$I^* = -2\log\left\{\sqrt{p_{\text{in}}p_{\text{out}}} + \sqrt{(1-p_{\text{in}})(1-p_{\text{out}})}\right\} \tag{3.7}$$

be the Bernoulli Rényi divergence of order $1/2$. For equal community sizes n/K,

$$\mathbb{E}\,\text{Mis}(\widehat{z}, z) \leq \frac{C}{\sqrt{np}}\exp\left\{-\frac{nI^*}{K}\right\}. \tag{3.8}$$

More generally, for $K \geq 3$ the size bound $n_a \geq n/(\beta K)$ gives

$$\mathbb{E}\,\text{Mis}(\widehat{z}, z) \leq \frac{C}{\sqrt{np}}\exp\left\{-\frac{nI^*}{\beta K}\right\}.$$

For $K = 2$, the bound is $C\exp\{-nI^*/2\}/\sqrt{np}$; its exponent does not involve β. These are consequences of the same Zhou–Li affinity bound, specialized to the planted-partition model.

Proof. For equal community sizes, the two differing blocks interchange p_{in} and p_{out}, so the Chernoff optimizer before removing the self-loop coordinate is $1/2$ and its information is nI^*/K. Removing one coordinate changes this information by $O(1)$ uniformly under the model conditions. In general, evaluating the Chernoff divergence at $1/2$ gives $D^*_{i,ab} \geq (n_a + n_b)I^*/2 - O(1)$. Apply Theorem 3.1, using $n_a + n_b \geq 2n/(\beta K)$, and $n_a + n_b = n$ when $K = 2$. The information I^* is the usual Rényi quantity in the community detection literature [5]; the displayed bounds retain the polynomial factor in the Zhou–Li oracle scale. □

3.2 Likelihood-based initialization supplies the required accuracy

For fitted profiles $\mathcal{C} = (\mu_1, \ldots, \mu_K)$ define

$$D(x\|y) = \sum_j \left\{x_j \log\frac{x_j}{y_j} + (1-x_j)\log\frac{1-x_j}{1-y_j}\right\}, \tag{3.9}$$

$$G(\mathcal{C}) = \sum_i \min_k D(q_i\|\mu_k), \qquad G_*(\mathcal{C}) = \sum_a n_a \min_k D(p_{a*}\|\mu_k),$$

$$\mathcal{F}(\widehat{\pi}, \mathcal{C}) = -\sum_i \log\sum_k \widehat{\pi}_k e^{-D(q_i\|\mu_k)}. \tag{3.10}$$

Writing $S_q = \sum_i \ell_i(q_i)$, the exact identities and inequalities

$$\mathcal{L} = S_q - \mathcal{F}, \qquad G \leq \mathcal{F}, \qquad \mathcal{F}(K^{-1}\mathbf{1}, \mathcal{C}) \leq G + n\log K \tag{3.11}$$

connect the algorithm's objective selection to a fitting-loss certificate.

Lemma 3.3 (Population replacement contraction)**.** *For every list of K fitted profiles, some replacement by a true profile satisfies*

$$G_*(\mathcal{C}^{r\leftarrow p_{a*}}) \le (1-1/K)G_*(\mathcal{C}). \tag{3.12}$$

Proof. Choose a minimizing fitted profile for each true class and a class contributing at least G_*/K. If the chosen-profile map is not onto, delete an unused profile; otherwise the map is a bijection and delete the chosen profile of this class. Inserting its true profile removes that class's loss, while all other classes retain a minimizing old profile. □

Theorem 3.4 (Empirical replacement gives a low-loss certificate)**.** *On the spectral event in Lemma 3.6, the replacement rounds of Algorithm 1 satisfy, uniformly in the round,*

$$\mathcal{F}_{t+1} \le (1-1/K)\mathcal{F}_t + Cn\sqrt{np} + n\log K, \tag{3.13}$$

$$\frac{\mathcal{F}_{T_n}}{n^2p} \le C(1-1/K)^{T_n} + \frac{C}{\sqrt{np}} + \frac{K\log K}{np} = o(1), \qquad T_n = \lceil \log n \rceil. \tag{3.14}$$

The statement holds for every starting fit in the common profile box and all simplex weights, including zero weights. A complete round without improvement also certifies $\mathcal{F} = o(n^2p)$. No trial is required to converge to a stationary point or a global optimum.

Lemma 3.5 (Low loss certifies the refined responsibility radius)**.** *On the same spectral event, consider any fit with exactly K profiles in the common box and $\mathcal{F} \le \xi_n n^2 p$, where $\xi_n \to 0$ is deterministic and uniform. After a common permutation, its profiles satisfy $\max_a \|\mu_a - p_{a*}\|_1 = o(np)$, its weights satisfy $-\log \min_a \widehat{\pi}_a = o(np)$, and its own exact E-step satisfies*

$$\frac{1}{n}\sum_{i,a} |R_{ia} - Z_{ia}| \le \frac{C}{np}. \tag{3.15}$$

These conclusions hold simultaneously for all such fits, including data-dependent fits.

The objective certificate is stronger than merely weak recovery. It makes the selected fit's existing final E-step accurate to $O((np)^{-1})$, so the mandatory final M-step already meets the sharp likelihood comparison below. Repeated replacement supplies this certificate regardless of the preceding growing trajectory. The proof does not assert general-K convergence of growing alone.

3.3 How much spectral compression changes the LRT

Let $M_0 = ZPZ^\top$, $M = \mathbb{E}[A \mid z]$, $\Pi = Z(Z^\top Z)^{-1}Z^\top$, and $\widehat{\Pi} = UU^\top$.

Lemma 3.6 (Spectral estimates with sufficient confidence)**.** *For every fixed $H > 0$, there is an event G_H with $\mathbb{P}(G_H^c) \le C_H e^{-Hnp}$ on which*

$$\|A - M\|_{\mathrm{op}} \le C_H\sqrt{np}, \qquad \|U\|_{2\to\infty} \le C_H/\sqrt{n}, \quad \|\widehat{\Pi} - \Pi\|_{\mathrm{op}} \le C_H/\sqrt{np}, \tag{3.16}$$

$$\|\widetilde{A} - M_0\|_{\mathrm{F}}^2 + \|q - M_0\|_{\mathrm{F}}^2 \le C_H np, \qquad \max_i \deg(i) \le C_H np, \qquad \kappa np/4 \le s \le 2np. \tag{3.17}$$

The retained eigenvalues have magnitudes of order np, and the true and observed profiles lie in a common box

$$[c_H p,\ \min\{C_H p, 1 - b_H\}]^n, \qquad b_H > 0. \tag{3.18}$$

Every true probability is a distance at least $c_H p$ from both clipping endpoints. Each graph obtained by deleting a specified vertex has the corresponding norm, eigenspace and row-norm bounds, with failure probability at most $C_H e^{-Hnp}$ and constants uniform in the deleted vertex.

The exponential confidence is essential at densities where $I_n \gg \log n$. An arbitrary fixed inverse-polynomial failure probability could dominate the target risk. Appendix A.2 verifies this confidence using Bandeira and van Handel [4, Corollary 3.12 and Remark 3.13] and Abbe et al. [2, Theorem 2.1 and Lemma 7].

The link to the oracle LRT uses a vertex-deleted graph only in the proof. Conditional on $B = A_{-i,-i}$, write $x = A_{-i,i}$ and let Π_B be the retained projector of B. A deterministic arrowhead expansion gives, for every bounded weight vector w,

$$\widetilde{A}_i w = x^\top \Pi_B w_{-i} + O(1) \tag{3.19}$$

on the degree and vertex-deleted spectral events. For a bounded block-constant w and every fixed bounded tilt parameter, the difference between the conditional log moment generating functions of $x^\top \Pi_B w_{-i}$ and $x^\top w_{-i}$ is $O(1)$. This is the relevant sense in which the lost edge-count information is small: at the testing boundary it changes the exponential tail by at most a constant factor. It need not make individual projected counts accurate to less than $1/2$.

Clipping and fitted parameters require further control. Once the responsibilities have error $O((np)^{-1})$, fitting changes the true-profile spectral score by at most $C(1+\mathcal{G})$, where $\mathcal{G}$ is a normalized global block fluctuation with bounded exponential moments. Clipping is handled by at most 2^K linear branches, rather than by claiming a uniform bounded difference between the final score and the raw LRT. Every branch that can cause an error has tilted variance of order np. Its probability in any interval of fixed length is $O((np)^{-1/2})$, which preserves the oracle prefactor as well as the Chernoff exponent.

Theorem 3.7 (Local refinement from weak responsibilities)**.** *Suppose a spectrum-based responsibility matrix has aligned average error at most a deterministic $\gamma_n \to 0$ outside an event of probability at most $C_H e^{-Hnp}$, for sufficiently large fixed H. One M-step, one E-step and a second M-step, followed by classification, give (3.5). If its aligned responsibility error is already at most $C/(np)$, a single M-step suffices. The conclusion remains valid after additional finite exact E/M updates and finite data-dependent stopping rules.*

For Algorithm 1 the low-loss certificate gives the stronger premise directly, so its stated final E/M update suffices. The proof in Appendices B and C.1 bounds the effect of compression on the LRT tail, including its $(np)^{-1/2}$ prefactor. This establishes (3.5) without recovering the integer edge counts.

3.4 Why the complete nonedge term matters

The sparse score used in earlier experiments is $\sum_j q_{ij} \log \widehat{p}_j - \sum_j \widehat{p}_j$. When $p \to 0$, its oracle contrast differs from the complete Bernoulli contrast by $O\{p \deg(i) + np^2\} = o(np)$ on the appropriate degree event. This approximation alone does not establish the Zhou–Li bound in (3.5).

At constant probabilities, dropping the nonedge term changes the testing rate. In the balanced model

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.2 & 0.2 \end{pmatrix},$$

only edges to the first block distinguish the hypotheses. For m such potential neighbors, the exact LRT threshold is $N/m = 1/2$, with information $0.22314355\, m$. The sparse score instead uses $N/m = 0.6/\log 4$, whose smaller directional error exponent is $0.13905314\, m$. Algorithm 1 retains the complete Bernoulli score throughout the density range.

4 The general degree-plus-spectrum algorithm

The preceding sections first treat the easier case in which the raw adjacency spectrum can be used directly. The general representation retains the degree vector as well as a regularized spectrum. Keeping the degrees supplies the constant direction in a likelihood contrast exactly; regularization addresses the influence of high-degree vertices on the estimated contrast directions. This section specifies that algorithm and the sharp testing comparison it requires. The exact identities and updates below do not require a logarithmic lower bound on the expected degree. For its statistical analysis we retain the balance, probability-comparability, full-rank, and upper-probability bounds in (2.2), with $p \le 1 - \eta$, and consider

$$np \longrightarrow \infty \tag{4.1}$$

in place of $np = \Omega(\log n)$. Theorem 3.1 remains the proved unconditional sharp guarantee for Algorithm 1 under its stated logarithmic-density assumption; its proof does not automatically establish the corresponding guarantee for the algorithm of this section.

4.1 Degree-preserving spectral compression

Write

$$\begin{gathered} D = A\mathbf{1}, \qquad \overline{D} = \frac{1}{n}\sum_i D_i, \qquad \tau = 2\overline{D}, \\ W = \operatorname{diag}(w_1, \dots, w_n), \qquad w_i = \begin{cases} \min\{1, \tau/D_i\}, & D_i > 0, \\ 1, & D_i = 0. \end{cases} \end{gathered} \tag{4.2}$$

Every diagonal entry of W is positive: a graph with an edge has $\overline{D} > 0$, and the all-zero graph has $W = I$. Form the softly degree-capped matrix $A_W = WAW$, and retain its K orthonormal eigenvectors with largest absolute eigenvalues, together with their signed eigenvalues:

$$A_W U = U\Lambda, \qquad U^\top U = I_K. \tag{4.3}$$

After this preprocessing, only (D, U, Λ) are retained. In particular, W is recoverable from D, and every subsequent operation uses these retained quantities.

Define the two $n \times (K+1)$ matrices

$$X = [\mathbf{1},\, WU], \qquad Y = [D,\, W^{-1}U\Lambda], \qquad R = YX^\dagger, \tag{4.4}$$

where $X^\dagger$ is the Moore–Penrose inverse. This definition allows a linearly dependent column, for example when the constant direction already belongs to $\operatorname{col}(WU)$. A rank-revealing factorization can be used in its computation. The matrix R is a reconstruction of the edge evidence, rather than an estimated matrix of independent edge probabilities.

Proposition 4.1 (Exact degree and block-count identities)**.** *For every adjacency matrix, the retained data in (4.4) determine*

$$AX = Y, \qquad R = A\Pi_X, \qquad R\mathbf{1} = D, \qquad \Pi_X = XX^\dagger. \tag{4.5}$$

For any candidate partition e, let $Z(e)$ be its membership matrix and set $\widetilde{b}(e) = RZ(e)$. Its rows obey

$$\sum_c \widetilde{b}_{ic}(e) = D_i, \qquad \widetilde{b}(e) - b(e) = -A(I - \Pi_X)Z(e), \qquad b(e) = AZ(e). \tag{4.6}$$

If $\operatorname{col}(Z(e)) \subseteq \operatorname{col}(X)$, then $\widetilde{b}(e) = b(e)$ exactly. In particular this conclusion applies to the true partition whenever its membership space is contained in the retained augmented space.

Proof. Multiplying (4.3) by W^{-1} gives $A(WU) = W^{-1}U\Lambda$, while $A\mathbf{1} = D$. Hence $AX = Y$ and $YX^\dagger = AXX^\dagger = A\Pi_X$. Since $\mathbf{1} \in \mathrm{col}(X)$, the orthogonal projector Π_X fixes $\mathbf{1}$, proving the degree identity. Multiplication by $Z(e)$ gives the count identity and its error formula. If $\Pi_X Z(e) = Z(e)$, the error is zero. □

The degree identity is exact even when the empirical subspace does not contain the true membership space. The separate condition for exact block counts is not asserted to hold for the empirical eigenvectors. In general, $\widetilde{b}_{ic}(e)$ is real-valued and may lie outside its binomial count range; no rounding guarantee follows. What the degree addition restores unconditionally is the sum of these counts, namely the observed degree. Also, $R = A\Pi_X$ need not be symmetric. Symmetrizing it would generally change its degree vector to $(D + \Pi_X D)/2$. Neither symmetrization nor entrywise clipping of R is part of the construction.

4.2 One Bernoulli working criterion for growing and refinement

Set

$$\varepsilon_D = \frac{c_0 \max\{1, \overline{D}\}}{n}, \qquad \widehat{p}_a \in [\varepsilon_D, 1 - \varepsilon_D]^n, \tag{4.7}$$

using the same admissible fixed c_0 as in (2.12). For $n \geq 2$, $\overline{D} \leq n - 1$ makes this interval nonempty. The complete Bernoulli working row score is

$$\ell_i^D(\widehat{p}) = \sum_{j=1}^{n} \{R_{ij} \log \widehat{p}_j + (1 - R_{ij}) \log(1 - \widehat{p}_j)\}. \tag{4.8}$$

Every logarithm is well-defined because the fitted profiles are bounded away from zero and one. The data coefficients R_{ij} themselves are not clipped. Thus (4.8) is an algebraic Bernoulli fitting criterion, not a probability density for signed observations. As with Algorithm 1, its reconstructed diagonal is retained; comparison to the loopless oracle must account for that coordinate in the score remainder.

For $v_j = \log\{\widehat{p}_j/(1 - \widehat{p}_j)\}$ and $\overline{v} = n^{-1} \sum_j v_j$, the degree contribution is explicit:

$$\ell_i^D(\widehat{p}) = D_i \overline{v} + R_i(v - \overline{v}\mathbf{1}) + \sum_j \log(1 - \widehat{p}_j). \tag{4.9}$$

The first term retains the total-degree evidence exactly. Only the centered contrast directions are approximated by the augmented spectral space. The last term keeps the nonedge contribution. Adding the degree vector does not by itself control the error in the centered contrasts at the oracle testing boundary.

Use the mixture objective and decoder in (2.14), with ℓ_i^D replacing ℓ_i. Candidate profiles and the initial global mean are, respectively,

$$\mathrm{clip}(R_c, \varepsilon_D, 1 - \varepsilon_D), \qquad \mathrm{clip}\left(\frac{1}{n}\sum_i R_i, \varepsilon_D, 1 - \varepsilon_D\right). \tag{4.10}$$

Growing and replacement use the same positive likelihood gains as in (2.15) and (2.18), evaluated with this score and these candidates. They require no L^2 or K-means initialization. The E-step and weight update are unchanged. To avoid confusing responsibilities with the reconstructed matrix R, denote the responsibilities here by r_{ia} and write $N_a = \sum_i r_{ia}$. The exact constrained profile M-step is

$$\widehat{p}_{aj}^{\,\mathrm{new}} = \mathrm{clip}\left(\frac{\sum_i r_{ia} R_{ij}}{N_a}, \varepsilon_D, 1 - \varepsilon_D\right), \qquad \widehat{\pi}_a^{\,\mathrm{new}} = N_a/n. \tag{4.11}$$

A zero-mass component retains its previous profile. This is an exact maximizer even when the weighted average is outside $[0,1]$: the derivative of its scalar objective has the sign of $\sum_i r_{ia} R_{ij}/N_a - \widehat{p}_{aj}$. Consequently each E/M pair increases the same working mixture objective. This algebraic monotonicity does not establish the replacement contraction or the statistical fitting certificate of Section 3; those require separate bounds for the new representation.

Algorithm 2: Degree-plus-spectrum likelihood decoding

1. Compute D, W, and the retained eigenpairs of WAW using (4.2)–(4.3). Retain only (D, U, Λ).
2. Form X, Y, and the degree-preserving reconstruction $R = YX^\dagger$. Keep R unchanged; constrain fitted and candidate profiles using (4.7)–(4.10).
3. Apply the global-mean growing, joint E/M fits, and $\lceil \log n \rceil$ replacement rounds of Algorithm 1, using (4.8) and (4.11) throughout.
4. Perform the mandatory final E-step and constrained profile/weight M-step, then decode by the fitted weighted row scores.

Relation to a center-based block-count decoder. The earlier degree-assisted coordinate construction can also be written using an invertible $K \times K$ matrix H of candidate block coordinates:

$$b^{(0)} = W^{-1} U \Lambda H^{-1}, \qquad b = b^{(0)} + (D - b^{(0)}\mathbf{1})\pi^\top, \qquad \pi^\top \mathbf{1} = 1.$$

For example, blockwise averages of the rows of WU supply candidate rows of H when that matrix is nonsingular, and estimated block proportions supply π. The correction makes $b\mathbf{1} = D$. It depends on the chosen coordinates and proportions; it is not asserted to equal the canonical projected counts $RZ(e)$ in (4.6). Algorithm 2 uses the latter reconstruction and the common free-profile row criterion, so no inverse of an estimated block-coordinate matrix is required.

4.3 The sharp testing comparison needed below logarithmic degree

The desired conclusion is the same Zhou–Li bound $\mathbb{E}\,\mathrm{Mis} \le C\eta^*$, rather than a different rate for the sparse case. Here η^* retains its exact definition in (3.3). The following sufficient comparison isolates the missing analytic task. It applies to a fitted decoder, including Algorithm 2 if its conditions can be verified, without imposing $np \gtrsim \log n$.

Proposition 4.2 (A sufficient sharp-transfer comparison)**.** *Assume (2.2), $p \le 1 - \eta$, and (4.1). Suppose the fitted components can be matched to the true labels by a common permutation. For each true label $a = z_i$ and each $b \ne a$, let $\widehat{L}_i^{ab}$ be their fitted score gap, including the fitted log-weight contrast, under this matching. Put $B = A_{-i,-i}$ and $x = A_{i,-i}$. All the conditions below are uniform in (i, a, b); their constants are fixed.*

(i) There is a B-measurable good event G and a possibly full-graph event E with $\mathbb{P}_a(G^c) + \mathbb{P}_a(E^c) \le C\eta^*$. On $G \cap E$,

$$\widehat{L}_i^{ab} = S_B(x) + \rho_B(x), \qquad S_B(x) = u(B) + \sum_{j \ne i} x_j v_j(B), \qquad |\rho_B(x)| \le h(B),$$

where $h \ge 0$, u, and v are B-measurable.

(ii) Let $\alpha = \alpha_{i,ab}^*$ be the oracle Chernoff optimizer. On G, the conditional incident-row log moment generating function satisfies

$$\psi_B(\alpha) = \log \mathbb{E}_a[e^{-\alpha S_B(x)} \mid B] \le -D_{i,ab}^* + c(B).$$

The conditional expectation here is over the full product-Bernoulli incident-row law, without conditioning on E.

(iii) Under the conditional tilted product law

$$Q_B(x) = \exp\{-\alpha S_B(x) - \psi_B(\alpha)\}\varphi_{i,a}(x),$$

one has, on G,

$$\sup_{t\in\mathbb{R}} Q_B\{S_B(x) \in [t, t+1]\} \le \frac{C}{\sqrt{np}}.$$

(iv) The remainder and log moment generating function cost obey $\mathbb{E}[\mathbf{1}_G \exp\{c(B) + \alpha h(B)\}] \le C$.

Then the decoder satisfies

$$\mathbb{E}\,\mathrm{Mis}(\widehat{z}, z) \le C\eta^*. \tag{4.12}$$

Proof. The uniform model bounds imply that α stays away from zero and one and that the oracle affinity comparison in (3.4) holds whenever $np \to \infty$; these oracle facts do not require uniform raw-adjacency concentration. On $G \cap E$, a wrong pairwise comparison requires $S_B \le h(B)$. After dropping the indicator of E, exponential tilting and a decomposition into the intervals $(h-k-1, h-k]$, $k = 0, 1, \ldots$, give

$$\begin{aligned}\mathbb{P}_a\{S_B \le h(B) \mid B\} &= e^{\psi_B(\alpha)}\mathbb{E}_{Q_B}[e^{\alpha S_B}\mathbf{1}\{S_B \le h(B)\}] \\ &\le \frac{Ce^{-D^*_{i,ab}+c(B)+\alpha h(B)}}{\sqrt{np}} \sum_{k=0}^{\infty} e^{-\alpha k}.\end{aligned}$$

The geometric sum is uniformly bounded. Integrating over $B \in G$, using (iv), and adding the exceptional probabilities proves a bound by $C\eta_{i,ab} + C\eta^* \le C\eta^*$ for the pairwise error. A union bound over the fixed number of competing labels and averaging over vertices prove (4.12). The same common matching can be used for all vertices; taking the minimum over permutations in Mis can only reduce the error. □

The proposition asks for an affine leave-one-out representation, control of its tilted log moment generating function, a local concentration bound, and an exponential-moment bound on the remaining score error. These conditions concern the retained evidence near the testing boundary; they do not assume the desired misclassification bound. Crucially, the degree-dependent matrix W, inverse reweighting, fitted profiles, and profile constraints must all be included when verifying them. The raw-spectrum proof and an earlier degree-calibration construction do not verify these conditions for the full regularized Algorithm 2 under (4.1). Thus this section restores the general degree-plus-spectrum algorithm and a precise sharp-rate target, while keeping the proved logarithmic-density theorem distinct from the remaining sparse-case analysis.

5 Experiments

The experiments address finite-sample decoding and initialization. They do not assume exact reconstruction of the integer community edge counts, and they do not estimate an asymptotic ratio to the oracle risk. We report two distinct numerical collections. A 60-graph benchmark evaluates the current full-Bernoulli Algorithm 1: global-mean growing followed by at most $\lceil \log n \rceil$ rounds of deterministic parameter replacement and a mandatory final M-step. The earlier 240-graph collection evaluates historical sparse-score procedures and is preserved under its original protocol.

The two collections use different scores, clipping rules, and fitting budgets; their results are reported separately. We also give an exactly certified population example explaining why a guarantee for growing alone cannot permit arbitrary finite intermediate EM budgets.

5.1 Current full-Bernoulli decoder: design and comparisons

The full-score design contains $n = 320$ vertices and $K \in \{2, 3, 4, 5, 6\}$, three matrix families, two density settings, and two independent graph seeds per cell, giving $5 \times 3 \times 2 \times 2 = 60$ graphs. The families are a balanced assortative model, an unequal-size disassortative model, and an unequal-size weak general model. Their block matrices are entrywise positive, symmetric, and full rank. The second family has negative signal eigenvalues; the third includes directions with weak finite-sample separation. The exact matrices, block sizes, and seeds are fixed in the saved protocol.

Each probability matrix is scaled so that the average expected degree, computed using the actual integer block sizes and excluding self-loops, equals either $3 \log n$ or $0.18(n-1)$. Expected degrees may differ between blocks. Both density settings use the same decoder and score; no adjacency regularization or edge-based refinement is applied.

Every method receives the same K raw adjacency eigenpairs with largest absolute eigenvalues. The full-score and sparse-score controls use exactly the same clipped reconstruction q. For this audit the clipping endpoints are ε and $1 - \varepsilon$, where

$$\varepsilon = 10^{-4} \frac{\max\{1, \max_a |\Lambda_{aa}|\}}{n}.$$

The reconstructed diagonal is retained. Thus the score-family comparison changes the working criterion while keeping the spectral information and clipping identical. It is not a comparison of graph likelihoods computed from different observations.

For each score family, the growing fit takes at most 100 EM updates at each stage, with unchanged labels and soft-loss change at most $10^{-9}n$ required for early stopping. Replacement tests every possible component removal, chooses its new node profile by the all-node likelihood gain, resets the trial weights uniformly, and takes one M-step. At most $\lceil \log 320 \rceil = 6$ rounds are allowed. The incumbent is retained unless a trial improves the same soft loss; if none improves it, repeating the unchanged deterministic round would give the same result. Both the growing baseline and the replacement decoder receive one final M-step before classification. The weight updates in this new audit are the unconstrained exact EM updates. These settings differ from the historical floored-weight and one-pass replacement controls below.

The Euclidean baselines run ten Lloyd starts on $\sqrt{n}U$ and $\sqrt{n}U\Lambda$, respectively, with at most 300 iterations per start. Ground truth is used to generate and evaluate the graphs; it is not passed to fitting, candidate ranking, likelihood selection, or stopping. Detailed entry points and data locations are given in Appendix D.

5.2 Current decoder: recovery results

Table 1 reports mean mislabeled fractions. Each family–density row averages ten graphs spanning the five values of K and two seeds. These are descriptive averages over the declared design, not repeated observations from one fixed parameter setting.

The full decoder has lower error than U k-means on 26 graphs, equal error on 27, and higher error on seven. Against $U\Lambda$ k-means, the corresponding counts are 21, 28, and 11. Against the

Table 1: A 60-graph comparison of the full-Bernoulli decoder. Entries are mislabeled fractions, not percentages. Within each score family, growing and replacement have identical errors on every graph. Each family–density row contains ten graphs.

Family	Density	Full score	Sparse score	U k-means	$U\Lambda$ k-means
Assortative	$3\log n$	0.008125	0.008125	0.008125	0.008125
Assortative	Dense	0	0	0	0
Unequal disassortative	$3\log n$	0.378750	0.377500	0.407813	0.391875
Unequal disassortative	Dense	0.074063	0.073438	0.095313	0.084063
Weak general	$3\log n$	0.663125	0.664375	0.662500	0.658438
Weak general	Dense	0.657500	0.659063	0.664063	0.660000
All 60 graphs		0.296927	0.297083	0.306302	0.300417

sparse-score control, they are eight, 43, and nine. Thus the small difference between the full and sparse aggregate errors does not establish a systematic finite-sample advantage of either score. The full criterion is used by the general-density algorithm because its likelihood contrasts admit the oracle rate comparison developed in the theory.

Replacement does not change the mislabeled fraction on any of the 60 graphs, for either score family. Seven trials are accepted by the strict floating-point loss comparison, but their apparent improvements are only 2.27×10^{-13} to 2.05×10^{-12}, with unchanged labels. They are rounding-level changes, not substantive empirical replacement improvements. This observation is consistent with a repair mechanism that is unnecessary once growing has reached a suitable fitted basin; it is not an initialization theorem for growing alone.

All decoders perform poorly on the weak general family. In particular, full rank and positive entries do not imply strong separation at $n = 320$. The archive includes the population signal eigenvalues and degree scales for these cases. None of the weak-signal graphs is removed from the summary. Increasing the density within this small design does not eliminate their large errors.

The audit records 15,444 EM updates. The largest positive soft-loss change is 2.73×10^{-12}; selected loss never increases across 127 evaluated replacement rounds. All 874 candidate scans retain every candidate loss and agree with the saved minimizing node ID. Every mandatory final M-step is executed; its largest positive loss change is 2.05×10^{-12}. These numerical checks agree with monotonicity to floating-point precision. The 60 adjacency matrices are distinct, symmetric, and have zero diagonal. Their saved eigenpairs have maximum residual $\|AU - U\Lambda\|_F = 1.08 \times 10^{-13}$ and maximum orthogonality error 4.53×10^{-15}. Direct calls to the fitting-time engine on two difficult graphs, using both score families, reproduce the archived labels exactly.

The implementation, fixed protocol, original graphs and probability matrices, retained eigenpairs, all fitted outputs, and complete candidate and EM traces are in

`experiments/20261010_general_density/`.

The files `benchmark_results/summary.json` and `benchmark_results/per_graph_methods.csv` contain aggregate and all 360 graph–method results. The protocol records the fitting-time code version and its hashes, alongside the data records; this new collection does not overwrite the earlier experiment.

5.3 A certified finite-stage population failure of growing

An explicit five-block example distinguishes convergence of an EM objective from a guarantee for arbitrary finite intermediate fitting budgets. For this population diagnostic only, write

$$P = \frac{d}{n}\begin{pmatrix} .0071 & .0914 & .0119 & .0129 & .0132 \\ .0914 & 2.7252 & .2538 & .2125 & .2722 \\ .0119 & .2538 & .0285 & .0276 & .0300 \\ .0129 & .2125 & .0276 & .0294 & .0289 \\ .0132 & .2722 & .0300 & .0289 & .0321 \end{pmatrix}, \qquad \pi = (.02, .57, .065, .22, .125). \tag{5.1}$$

Here $d \to \infty$ controls the probability scale of this example, with $d/n \to 0$; it introduces no additional assumption on the main model and does not require equal block degrees. The displayed shape matrix is positive definite: the leading principal determinants after multiplying it by 10^4 are exactly 71, 1099496, 22194480, 49511806, and 22302154.

Use population block profiles, the archived sparse working score, the exact weight update with floor 10^{-8}, and one M-step after each added center. With block indices starting at one, the growing procedure selects blocks $4, 1, 5, 4$. Its limiting centers are

$$(p_{2*}, p_{4*}, p_{1*}, m_{35}, p_{4*}), \qquad m_{35} = \frac{\pi_3 p_{3*} + \pi_5 p_{5*}}{\pi_3 + \pi_5}.$$

The fourth center merges blocks 3 and 5, while two centers duplicate block 4. Consequently the limiting mislabeled fraction is $13/200 = 0.065$. After the penultimate M-step, block 3 has changed its preferred component, but the previous mixed center still contains it. The next global gain then selects block 4 again. The old mixed component has exponentially small responsibilities whose normalized M-step is itself dominated by block 4; setting these responsibilities to numerical zero is not needed for the failure. Excluding selected node IDs also does not exclude other nodes of the same block.

The limiting candidate comparisons and assignments are certified with exact rational intervals for logarithms, using an explicit series remainder. The smallest candidate-selection margin is greater than 7.98×10^{-7}; the vanishing component's soft M-step has a unique dominating block with margin greater than 7.58×10^{-4}. Numerical calculations that normalize the M-step in log space reproduce error 0.065 for $d = 10^5, 10^6, 10^8, 10^{10}, 10^{12}$, including 100 additional final EM updates. One deterministic replacement round, with one M-step per proposal, removes a duplicate center and inserts block 3, reducing the error to zero. At $d = 10^6$, the centered per-node working log likelihood rises from -17.505006 to -1.169355.

This is a population counterexample to a guarantee allowing just one intermediate M-step. It does not establish failure when every growing stage is fitted to convergence: two M-steps per stage succeed in this same example. It is also not a sampled-graph failure theorem. The complete matrix, strict comparison certificates, score traces, and successful replacement trials are in `experiments/20261010_general_density/population_audit/`. The general-K guarantee for Algorithm 1 instead uses the deterministic replacement argument in Theorem 3.1, without assuming that growing alone has already found every community.

5.4 Historical sparse-score collection: models and protocol

The earlier 240-graph collection uses the sparse working score and its original fitting settings. It is retained as historical evidence; these graphs were not rerun with the new full-Bernoulli Algorithm 1.

For this archived design, let $C^{(0)}$ be a fixed symmetric connectivity matrix and let π be the vector of community proportions. We generate an undirected SBM without self-loops, with edge probabilities

$$P_{ab} = \frac{\log}{n} n C_{ab}, \qquad C = \frac{\eta}{D_{\min}(C^{(0)}, \pi)} C^{(0)}, \tag{5.2}$$

where $\eta \in \{0.8, 1.4\}$ and

$$D_{\min}(C, \pi) = \min_{a \neq b} \max_{0 \leq t \leq 1} \sum_{c=1}^{K} \pi_c \left[(1-t) C_{ac} + t C_{bc} - C_{ac}^{1-t} C_{bc}^{t} \right]. \tag{5.3}$$

Thus the limiting minimum Chernoff–Hellinger coefficient equals η. The normalization uses the limiting proportions π; actual group sizes are $\lfloor n\pi_a \rfloor$, with the remaining vertices assigned to the last group. Vertex labels are then randomly permuted. Each model is tested at $n \in \{1000, 2000\}$. The six models are

$$C^{(0)}_{\text{sym2}} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}, \qquad \pi_{\text{sym2}} = (1/2, 1/2),$$

$$C^{(0)}_{\text{equal3}} = \begin{pmatrix} 10 & 1 & 2 \\ 1 & 11 & 1 \\ 2 & 1 & 10 \end{pmatrix}, \qquad \pi_{\text{equal3}} = (1/3, 1/3, 1/3),$$

$$C^{(0)}_{\text{hetero3}} = \begin{pmatrix} 6 & 0.5 & 0.5 \\ 0.5 & 12 & 0.5 \\ 0.5 & 0.5 & 24 \end{pmatrix}, \qquad \pi_{\text{hetero3}} = (1/3, 1/3, 1/3),$$

$$C^{(0)}_{\text{unequal3}} = \begin{pmatrix} 10 & 1 & 1 \\ 1 & 10 & 1 \\ 1 & 1 & 10 \end{pmatrix}, \qquad \pi_{\text{unequal3}} = (0.5, 0.3, 0.2),$$

$$C^{(0)}_{\text{hier4}} = \begin{pmatrix} 12 & 2 & 0.5 & 0.5 \\ 2 & 12 & 0.5 & 0.5 \\ 0.5 & 0.5 & 12 & 2 \\ 0.5 & 0.5 & 2 & 12 \end{pmatrix}, \qquad \pi_{\text{hier4}} = (1/4, 1/4, 1/4, 1/4),$$

$$C^{(0)}_{\text{dis3}} = \begin{pmatrix} 0.5 & 8 & 8 \\ 8 & 0.5 & 8 \\ 8 & 8 & 0.5 \end{pmatrix}, \qquad \pi_{\text{dis3}} = (1/3, 1/3, 1/3).$$

These include unequal group sizes, heterogeneous expected degrees, hierarchical connections, and negative signal eigenvalues.

The design batch contains five graph seeds, $20000, \ldots, 20004$, for every model, size, and signal coefficient, giving $6 \times 2 \times 2 \times 5 = 120$ graphs. A second batch uses the same models and hyperparameters with fresh seeds $30000, \ldots, 30004$. The algorithms and second-batch seed range were fixed before fitting the second batch. The second batch therefore validates initialization choices made using the first batch, rather than introducing new models. All methods are evaluated on identical eigenpairs within a graph.

We retain the K eigenpairs of the raw adjacency matrix with largest absolute eigenvalues. The implementation requests $K + 1$ eigenpairs using the symmetric ARPACK solver with tolerance

10^{-8} and discards the last one after sorting by absolute value. After this preprocessing, the fitting routines receive only (U, Λ); adjacency entries, degrees, true labels, and true block probabilities are unavailable to fitting and model selection.

5.5 Historical sparse-score implementations and comparisons

The archived implementation forms positive spectral profiles

$$q_{ij} = \max\left\{(U\Lambda U^\top)_{ij}, 10^{-4}\frac{\log}{n}n\right\}, \qquad \ell_i^{\mathrm{sp}}(\widehat{p}) = \sum_{j=1}^{n} q_{ij}\log\widehat{p}_j - \sum_{j=1}^{n}\widehat{p}_j. \tag{5.4}$$

The reconstructed diagonal is retained, even though the generated graphs have zero diagonal. Thus this is a declared positive-profile working score, not the original Bernoulli graph likelihood. Component means are unrestricted n-dimensional profile means. Every likelihood arm in this historical collection uses the same soft EM updates and observed working objective

$$\mathcal{L}_{\mathrm{sp}}(\widehat{\pi}, \widehat{p}) = \sum_{i=1}^{n}\log\left[\sum_{a=1}^{K}\widehat{\pi}_a \exp\{\ell_i^{\mathrm{sp}}(\widehat{p}_a)\}\right]. \tag{5.5}$$

EM initializes equal component weights except when refitting a replacement proposal, where the current weights are retained. The mean update is the responsibility-weighted average of q_i. The weight update exactly maximizes the EM auxiliary objective on $\{\widehat{\pi}_a \geq 10^{-8},\ \sum_a \widehat{\pi}_a = 1\}$. No additional floor on means is needed because the observations are positive. Final labels maximize $\log\widehat{\pi}_a + \ell_i^{\mathrm{sp}}(\widehat{p}_a)$.

The historical growing arm uses the global-mean and all-node-gain structure of Section 2.5, with the archived sparse score and clipping rule above. It is recorded as `growing_global_gain_EM`. It starts from the global profile mean, adds $K-1$ node candidates by their all-node marginal gains, and runs joint EM after each addition. It requires no top-subset size and removes no nodes.

The frozen-profile control, recorded as `global_gain_no_update`, starts with the historical top-subset rule and then chooses all remaining node seeds by all-node gains, keeping the profiles frozen until one final EM fit. Its exact historical subset sizes are recorded in Appendix D. It is a different algorithm despite its identical recovery errors in these experiments.

We compare it with the following alternatives. *Top-subset peeling* uses the historical first-seed rule and repeatedly removes the selected candidate and its best-scoring subset, using the archived sizes specified in the appendix. After K selections, all n observations, including unselected nodes, enter EM. *Peeling plus replacement* fits this initialization, deletes each component temporarily, proposes the globally most useful node against the remaining $K-1$ means, and refits all nodes. All K proposals use the same original fitted endpoint; the best proposal is accepted only if its final $\mathcal{L}_{\mathrm{sp}}$ exceeds the current best by more than 10^{-5}. This is one replacement pass, not repeated optimization.

In the historical API, `method='global_gain'` invokes the growing arm and does not invoke the frozen-profile control. The following randomized procedures are archived comparison arms, not steps of the current Algorithm 1. *Residual++* samples the first node uniformly and later nodes proportionally to

$$d_i = \min_{a\in S}\sum_j \left[q_{ij}\log(q_{ij}/\widehat{p}_{aj}) - q_{ij} + \widehat{p}_{aj}\right]. \tag{5.6}$$

This deviance is not squared again. Its greedy variant samples $2 + \lfloor \log K \rfloor$ candidate trials per addition and chooses the trial that minimizes total residual deviance. We report both one start

Table 2: Average mislabeled fraction in percent; exact recovery counts are out of 120 graphs per batch. Historical sparse-score arms only; this table does not evaluate the current Algorithm 1.

	Design batch		Independent batch	
Method	Error (%)	Exact	Error (%)	Exact
Top-subset peeling + EM	0.3917	86	0.3725	77
Peeling + replacement	**0.0967**	86	**0.0925**	77
Frozen global gain + EM	0.0967	86	0.0925	77
Growing global gain	**0.0967**	86	**0.0925**	77
Growing sampled gain + EM	0.3900	86	0.0925	77
Residual++, one start + EM	4.0129	80	3.1675	72
Greedy residual++, one start + EM	1.1192	85	2.3438	74
Greedy residual++, ten starts + EM	0.0967	86	0.0925	77
Euclidean++, one start + EM	3.9033	78	6.4204	71
X k-means, ten starts	0.3454	73	0.2933	66

and ten starts; the ten-start fit is selected solely by final $\mathcal{L}_{\mathrm{sp}}$. An additional control selects one set of Euclidean k-means++ seeds in $X = \sqrt{n}U\Lambda$ and then runs the same profile EM. The main Euclidean baseline runs k-means in X with ten starts, at most 300 Lloyd iterations per start, and tolerance 10^{-5}.

All profile EM fits use at most 300 updates and tolerance 10^{-6}, with label, parameter-change and objective-change checks. The exact archived stopping rule is recorded in Appendix D. Randomized seeding uses fixed NumPy seed sequences indexed by graph seed, method, and restart. Reported error rates minimize label disagreement over all permutations of the K labels. Averages weight graphs equally; "exact" counts graphs with no mislabeled node. True labels are accessed only after all fitting and likelihood-based selections finish.

5.6 Historical sparse-score recovery results

Table 2 gives the paired results in the design and independent batches. Global-gain initialization, peeling plus replacement, growing global gain, and ten-start greedy residual++ have the same per-graph error counts in both batches. Across all 240 graphs, the archived growing arm has average error 0.0946%, compared with 0.3194% for the ten-start Euclidean baseline, and exactly recovers 163 graphs compared with 139. These are finite-sample recovery comparisons with unequal optimization budgets, rather than equal-runtime comparisons.

Table 3 separates model families. The largest improvement over Euclidean decoding occurs with heterogeneous diagonal connectivities: the average error falls from 1.4175% to 0.1625%. The likelihood criterion also improves the unequal-size model. It is not uniformly better: the symmetric two-community baseline has one fewer mislabeled node in aggregate, and the hierarchical model has one additional exactly recovered graph under k-means. The examples therefore support likelihood adaptation when block profiles differ substantially, without claiming dominance on every graph.

The stronger signal coefficient $\eta = 1.4$ gives 117 exactly recovered graphs out of 120 for global gain, versus 99 for k-means. At $\eta = 0.8$, the corresponding counts are 46 and 40. This behavior is consistent with the importance of the logarithmic-density recovery threshold, but the experiment does not establish that threshold or the leading error exponent.

Table 3: Both batches combined, 40 graphs per model. The likelihood columns use the archived sparse-score growing branch. The frozen-profile control has identical recorded error counts.

	Growing global gain + EM		X k-means	
Model	Error (%)	Exact	Error (%)	Exact
Symmetric two-community	0.0150	33	0.0125	34
Equal-size three-community	0.0175	32	0.0188	31
Heterogeneous three-community	0.1625	18	1.4175	1
Unequal-size three-community	0.0338	33	0.1213	25
Hierarchical four-community	0.0275	28	0.0288	29
Disassortative three-community	0.3113	19	0.3175	19

5.7 Historical subset-score failure and its repair

One design graph with $n = 2000$, $K = 3$, and $\eta = 0.8$ illustrates the initialization mechanism. Its normalized connectivity matrix is approximately

$$C = \begin{pmatrix} 7.8541 & 0.7854 & 1.5708 \\ 0.7854 & 8.6395 & 0.7854 \\ 1.5708 & 0.7854 & 7.8541 \end{pmatrix}, \tag{5.7}$$

with group sizes $(666, 666, 668)$. Communities 1 and 3 are the closest pair: their empirical X-centroid distance is 40.22, compared with 47.36 and 47.82 for the other pairs. They are nevertheless well separated in the retained spectrum. Using their true profile means as a diagnostic classifier assigns every node correctly on its first assignment.

Top-subset peeling selects nodes $858, 610, 435$ (zero-based IDs), whose true communities are $2, 3, 2$. The profile of node 610 lies near the mixture of communities 1 and 3. Its profile divergence to their merged mean is 0.3826, compared with 5.2433 to its own community's mean. Its selected subset has 150 nodes from community 1 and 196 from community 3. Thus a candidate can win a subset likelihood comparison by representing a mixture of communities. The resulting fit merges communities 1 and 3 and splits community 2, making 709 errors (35.45%). Tightening the tolerance to 10^{-10} and permitting 3000 updates does not solve this structural problem: the fit stops after 517 updates with 800 errors.

The frozen global-gain control keeps the first two seeds but changes the third to node 220, which contributes to the uncovered region; after EM, only one node is mislabeled. A one-component replacement at the original fitted endpoint selects the same node and likewise reduces errors to one, while increasing the working log likelihood by approximately 7148.48. No true labels enter either choice. The independent batch contains a second analogous peeling failure with 675 errors, reduced to three by replacement. Relative to peeling, each effective deterministic variant improves exactly one graph in each batch, leaves the other 119 error counts unchanged, and worsens none. The number of exact recoveries does not change because the repaired graphs retain one and three errors, respectively.

5.8 Historical numerical verification and computational scope

All returned historical profile fits satisfy the declared stopping rule in the 240-graph comparison. The 120 original peeling errors are reproduced exactly. The archive contains 6240 EM traces, including replacement proposals, growing stages, and restart fits, with 27867 recorded trace points. The

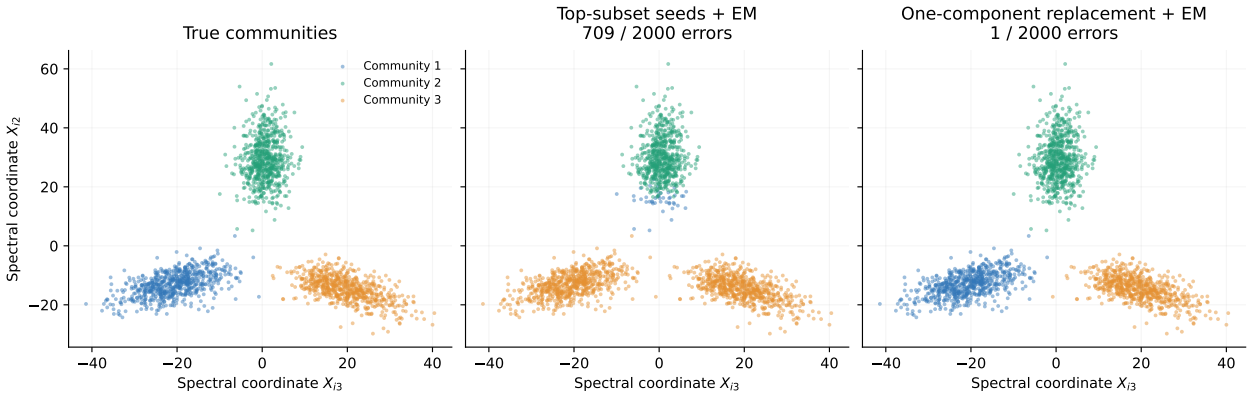


Figure 1: The saved design failure, displayed in two coordinates of $X = \sqrt{n}U\Lambda$. Left: true communities, used only for evaluation. Middle: top-subset peeling followed by EM merges the two lower communities and splits the upper community. Right: an accepted one-component replacement followed by EM separates the groups and leaves one mislabeled node. Predicted labels are aligned to truth for display only.

smallest observed objective increment is -1.17×10^{-10}, within floating-point rounding; no increment below -10^{-7} occurs. All 240 selected ten-start solutions maximize the same final working likelihood within their restart set. Replacement accepts a likelihood improvement on 22 design graphs and 25 independent graphs, and never lowers the final objective; most of these small improvements do not change the classification error. Independent scalar checks reproduce deviance and global-gain calculations with maximum absolute errors 7.11×10^{-15} and 1.78×10^{-15}, respectively.

The present implementation explicitly reconstructs the $n \times n$ profiles and caches all singleton-candidate scores. Reconstruction costs $O(n^2K)$, score precomputation costs $O(n^3)$, and storage costs $O(n^2)$. An EM update costs $O(n^2K)$. A replacement pass tries K additional fits, whereas ten-start residual seeding uses ten fits. The runs use one numerical library thread per worker and four worker processes. Python, NumPy, SciPy, and scikit-learn versions are 3.11.7, 2.1.3, 1.16.2, and 1.7.1. Stored timers cover all candidate arms and restarts together and exclude graph generation and eigenpair computation, so they do not support an arm-specific speed claim. The data sizes $n \leq 2000$ establish feasibility for this implementation; candidate-pool approximations and blockwise computations require separate large-graph validation.

The archived comparisons support improved decoding of a fixed spectrum in several heterogeneous models and identify a concrete failure of subset initialization. The new 60-graph study directly evaluates the current Algorithm 1 and records its weak-signal failures and its lack of a substantive replacement improvement on that particular collection. Neither collection estimates an asymptotic error exponent, proves uniform dominance over k-means, or replaces the model assumptions and analysis of Theorems 3.1 and 3.7.

Companion sparse threshold-certificate checks. A separately retained sparse-score implementation tests a uniform-weight refit together with the replacement trials and accepts only gains of at least $(\sum_{ij} q_{ij})/\sqrt{\log n}$. Its archived checks contain 24 graphs with $n \in \{256, 512\}$, $K \in \{3, 4, 6\}$, two degree scales, and two seeds in one heterogeneous assortative family. They record 19 exact recoveries and seven errors among 9,216 nodes; no natural growing endpoint requires an accepted replacement in that collection. A separate injected collapsed population endpoint is repaired in two accepted

rounds. These checks concern the companion sparse threshold rule, not the complete Bernoulli Algorithm 1. They remain available as `general_k_fresh_checks.json` beside that implementation.

5.9 Conditional tilted-row diagnostic

The proof compares spectral contrasts with the oracle LRT near the classification boundary, rather than demanding exact integer counts. A separate diagnostic checks its arrowhead reduction on 144 graph–row configurations: three model scenarios (dense assortative, dense disassortative, and a sparse probability scale $8 \log n/n$), four sizes $n \in \{200, 400, 800, 1600\}$, and twelve seeds per cell. For each configuration, the vertex-deleted graph is sampled from the original model and its incident row is sampled from the corresponding oracle Chernoff-tilted law. This makes near-boundary observations common. Seeds are reused across design cells, so these configurations are not asserted to be mutually independent across cells.

For the bounded oracle weight used in the check, the largest observed absolute arrowhead remainder was 2.063, and the exact algebraic identity had numerical error at most 7.3×10^{-10}. The normalized subspace and Gram-matrix errors stayed bounded over the tested sizes. These results check the algebra and the constant-scale comparison used in the proof. They do not prove a rare-event probability bound, exact recovery of integer counts, or equality of spectral and oracle error probabilities. The executable diagnostic and all saved results are in

`experiments/20261018_sharp_rate/`.

6 Conclusion

An oracle SBM classifier uses community edge counts through their Bernoulli likelihood contrasts. Our first theorem controls the loss caused by an empirical unregularized spectral reconstruction: for $p = \Omega(\log n/n)$ and the stated model conditions,

$$\mathbb{E}\ \mathrm{Mis}(\widehat{z}, z) \le C\eta^*.$$

The proof controls likelihood tails after projection, clipping and fitting, including their local concentration under the oracle exponential tilt. The general-K replacement certificate and the algorithm's existing final E/M update supply the required parameter accuracy. This result permits unequal community sizes and expected degrees, intermediate and dense probabilities, and either sign of signal eigenvalues. It requires neither integer count recovery nor a risk ratio tending to one.

The general algorithm retains both observed degrees and the eigenpairs of WAW. The identities $A\mathbf{1} = D$ and $A(WU) = W^{-1}U\Lambda$ permit exact evaluation of A on their augmented span. Its reconstructed block counts preserve every observed degree, and initialization and refinement use one complete Bernoulli row criterion. This construction is defined below logarithmic expected degree. The general transfer proposition preserves the same Zhou–Li bound once its conditional cumulant, local concentration and fitted-score hypotheses hold. Establishing those hypotheses uniformly for the full regularized algorithm when $np \to \infty$ is the remaining sparse-case proof task; the unconditional theorem of the earlier sections is confined to its stated density range.

The numerical results concern the unregularized implementations and illustrate finite-sample behavior, including weak-signal errors and the absence of a classification gain from replacement in the complete 60-graph collection. The general implementation is checked for its exact retained-information identities and EM objective updates. These checks do not establish a sparse statistical

rate. Further questions include reducing the cubic candidate-cache cost and quadratic storage, relaxing the rank condition, and obtaining sharp affinity comparisons at vanishing block probabilities.

A Oracle testing and spectral estimates

All probabilities and expectations below are under the SBM with the fixed parameter z. Set $C_a = \{j : z_j = a\}$, $M_0 = ZPZ^\top$, $M = \mathbb{E}[A \mid z]$, and $\Pi = Z(Z^\top Z)^{-1}Z^\top$. The empirical projector is $\widehat{\Pi} = UU^\top$. In the proofs μ_a denotes a fitted profile $\widehat{p}_a$. Constants may depend on the fixed model bounds, K, c_0, the density lower-bound constant and a fixed confidence parameter H, but not on n or a normalized matrix limit. We sometimes write $d = np$ within an argument.

A.1 Uniform information and the oracle LRT rate

For the fixed pair a, b define blockwise weights and nonedge contributions

$$w_c^{ab} = \log \frac{P_{ac}(1 - P_{bc})}{P_{bc}(1 - P_{ac})}, \qquad r_c^{ab} = \log \frac{1 - P_{ac}}{1 - P_{bc}}.$$

Then the LRT in (2.6) is $L_i^{ab} = \sum_{j \neq i}\{A_{ij} w_{z_j}^{ab} + r_{z_j}^{ab}\}$. We write $\alpha_{i,ab}^*$ for the minimizer of $\mathbb{E}_a \exp(-\alpha L_i^{ab})$, so its minimum equals $e^{-D_{i,ab}^*}$.

Lemma A.1 (Uniform oracle constants)**.** *The bounds $D_{i,ab}^* \asymp np$, $\alpha_{i,ab}^* \in [\alpha_0, 1 - \alpha_0]$, and tilted variance $\asymp np$ hold uniformly over (2.2)–(2.3). The equal-prior binary oracle error is between constant multiples of $e^{-D_{i,ab}^*}/\sqrt{np}$.*

Proof. The odds weights are uniformly bounded and, for $u, v \in [\kappa p, p]$ with $p \leq 1 - \eta$, their absolute difference in logits is between $|u - v|/p$ and $|u - v|/(\kappa\eta p)$. Distinct rows of P are separated in Euclidean norm by at least $\sqrt{2}\sigma_{\min}(P)$. Positive community proportions imply

$$\min_{a \neq b} \|p_{a*} - p_{b*}\|_1 \geq cnp, \qquad cnp \leq D(p_{a*} \| p_{b*}) \leq Cnp. \tag{A.1}$$

Indeed, $\|p_{a*} - p_{b*}\|_2^2 \geq cnp^2$ and $\|p_{a*} - p_{b*}\|_\infty \leq p$ give the first inequality. Bernoulli entropy curvature at least $1/p$ on the interval between two true probabilities gives the KL lower bound; its upper bound follows from probability comparability and $1 - p \geq \eta$. Thus $\sum_c (n_c - \mathbf{1}\{z_i = c\})(w_c^{ab})^2 \asymp n$. At any tilt $0 \leq \alpha \leq 1$, its Bernoulli probabilities lie between P_{ac} and P_{bc}. Their variances are therefore between constant multiples of p, so the second derivative of $\psi(\alpha) = \log \mathbb{E}_a e^{-\alpha L_i^{ab}}$ is between $c_1 np$ and $C_1 np$. Since $\psi(0) = \psi(1) = 0$, integrating this curvature gives information $\asymp np$ and an interior optimizer bounded away from both endpoints. Bounded log-odds weights imply that the sum of tilted third absolute centered moments is at most a constant times the tilted variance. The rate-optimal Chernoff bounds of Zhou and Li [12] then give the two-sided oracle rate. □

A.2 Exponential-confidence spectral bounds

Proof of Lemma 3.6. The actual mean M leaves the membership space invariant. In its orthonormal membership basis its restriction is

$$(Z^\top Z)^{1/2} P (Z^\top Z)^{1/2} - \operatorname{diag}(P_{11}, \ldots, P_{KK});$$

on the orthogonal complement its eigenvalues are the numbers $-P_{aa}$. Consequently the K signal eigenvalues have magnitudes between $c_1 np$ and $C_1 np$, their projector is exactly Π, and their population row norms are $O(n^{-1/2})$.

The bounded-entry matrix norm inequality of Bandeira and van Handel [4, Corollary 3.12 and Remark 3.13] (for centered, not necessarily symmetrically distributed entries) gives, after absorbing $\sqrt{\log n}$ into $\sqrt{np}$,

$$\mathbb{P}\{\|A - M\|_{\mathrm{op}} > C\sqrt{np} + t\} \le n e^{-t^2/C}.$$

Taking $t = C_H\sqrt{np}$ provides any prescribed fixed confidence e^{-Hnp}, since $\log n \le c^{-1} np$. A degree Chernoff bound with a sufficiently large fixed multiplier gives the maximum-degree assertion at the same confidence.

For completeness, we verify the confidence parameter in the entrywise eigenspace theorem of Abbe et al. [2], rather than replacing an exponential confidence by a polynomial one. In their Theorem 2.1 use

$$\gamma = \frac{C_H}{\sqrt{np}}, \qquad \varphi_H(x) = \frac{C_H}{1 \vee \log(1/x)}\ (x > 0), \qquad \varphi_H(0) = 0, \quad \delta_0, \delta_1 \le e^{-Hnp}.$$

Their Bernoulli row inequality (Lemma 7) holds for every fixed nonnegative row-tail parameter. Choose that parameter sufficiently larger than $H + c^{-1} + 1$, where $np \ge c \log n$. It gives per-row failure at most δ_1/n. Applied to each of the fixed number of columns, the monotonicity of φ_H and of $\varphi_H(x)/x$ verifies their matrix row-concentration assumption. Factors depending on K and the signal gap are included in C_H. Row/column independence follows from the independent upper-triangular edges. Their mean-row condition follows from $\|M\|_{2\to\infty} \le np/\sqrt{n}$ and $np \le n$. Their smallness condition holds eventually because $\gamma \to 0$ and $\varphi_H(\gamma) \to 0$.

Apply the theorem separately to the positive and negative signal groups, each separated by a gap of order np from its complement. Its row-norm bound gives $\|U\|_{2\to\infty} \le C_H/\sqrt{n}$. The theorem does not require M to have exactly rank K, so the no-loop diagonal correction is retained. The operator norm event and the signal gap give the asserted projector bound by the usual subspace perturbation inequality. A fixed number of union bounds is absorbed by increasing the confidence parameter in the preceding steps.

The Perron eigenvalue of M lies between its minimum and maximum row sums, which are in $[\kappa(n-1)p, np]$. The norm event gives $\kappa np/4 \le s \le 2np$ eventually. Thus $\varepsilon_n \asymp p$, every true probability is at distance at least $c'p$ from both clipping endpoints, and

$$|\widetilde{A}_{ij}| \le \|U_i\|_2 \|\Lambda\|_{\mathrm{op}} \|U_j\|_2 \le C_H p.$$

This proves the common box: if $C_H p \le 1/2$ it is bounded away from one directly; otherwise $\varepsilon_n \ge c_H p \ge c_H/(2C_H)$ gives a fixed upper gap. Removing one vertex preserves all model bounds with adjusted fixed constants, and the same argument proves the vertex-deleted assertions uniformly in i. In a per-node risk calculation one only needs the event for that particular deleted vertex. Finally, best rank-K approximation in operator norm gives

$$\|A - \widetilde{A}\|_{\mathrm{op}} \le \|A - M_0\|_{\mathrm{op}} \le C_H\sqrt{np}.$$

Since $\widetilde{A} - M_0$ has rank at most $2K$, this implies $\|\widetilde{A} - M_0\|_{\mathrm{F}}^2 \le C_H np$. Clipping toward an interval containing every entry of M_0 contracts coordinatewise squared error, and hence gives the stated bound for q as well. □

B Likelihood initialization and the refined responsibility radius

B.1 Exact EM and the profile constraint

For any nonnegative responsibilities with rows summing to one, define

$$\mathcal{Q}(R, \widehat{\pi}, \widehat{p}) = \sum_{i,a} R_{ia}\{\log \widehat{\pi}_a + \ell_i(\widehat{p}_a) - \log R_{ia}\}.$$

The log-sum inequality gives $\mathcal{Q} \le \mathcal{L}$, with equality at the E-step. For fixed R, the profile-coordinate objective is

$$\left(\sum_i R_{ia} q_{ij}\right) \log \widehat{p}_{aj} + \left(N_a - \sum_i R_{ia} q_{ij}\right) \log(1 - \widehat{p}_{aj}).$$

Its maximizer for $N_a > 0$ is the weighted mean in (2.17); maximizing $\sum_a N_a \log \widehat{\pi}_a$ over the simplex gives $\widehat{\pi}_a = N_a/n$. Thus every exact E/M pair raises $\mathcal{L}$ and lowers $\mathcal{F}$. A zero-mass component contributes nothing to this objective and may retain its previous profile. All nonzero-mass updates are convex combinations of the q rows. These observations also prove that all means remain in the common box on G_H.

Since $D(q_i \| \widehat{p}_a) \ge 0$, the objective is bounded above by the observation-only saturated score S_q. Consequently an infinite exact EM sequence has convergent objective values. This fact alone gives neither a global optimum nor community recovery. The replacement argument below supplies the latter conclusion from a different, finite-round bound.

B.2 Replacement contracts the common fitting loss

Proof of Theorem 3.4. Write $d = np$. The Frobenius bound gives $\sum_i \|q_i - p_{z_i,*}\|_1 \le n\|q - M_0\|_{\mathrm{F}} \le Cn\sqrt{d}$. Define

$$G(\mathcal{C}) = \sum_i \min_k D(q_i \| \mu_k), \qquad G_*(\mathcal{C}) = \sum_a n_a \min_k D(p_{a*} \| \mu_k).$$

The scalar divergence derivatives are

$$\partial_x D(x\|y) = \log \frac{x(1-y)}{y(1-x)}, \qquad \partial_y D(x\|y) = \frac{y - x}{y(1-y)}.$$

Both are uniformly bounded on the common box. Consequently the profile divergence is uniformly Lipschitz in each argument's ℓ_1 norm, so $|G - G_*| \le Cn\sqrt{d}$ simultaneously over every admissible center list. Each true class supplies an observed candidate within $C\sqrt{d}$ in ℓ_1 of its true profile.

For any K fitted centers, choose a minimizing center for each true class. A class contributing at least G_*/K exists. If the chosen-center map is not onto, delete an unused center; otherwise it is a bijection, and delete the center chosen by this class. Inserting its true profile removes at least that class's contribution and harms none of the other classes. Thus some ideal replacement decreases population hard loss by at least G_*/K. Replacing the ideal profile by its available observed candidate and applying the uniform Lipschitz bounds gives an observed trial with hard loss at most $(1 - 1/K)G + Cn\sqrt{d}$.

For each deletion r, the saturation identity $\ell_i(\mu) = \ell_i(q_i) - D(q_i \| \mu)$ gives

$$G(\mathcal{C}^{r \leftarrow q_c}) = G(\mathcal{C} \setminus \{\mu_r\}) - \sum_i (\ell_i(q_c) - h_i^{(-r)})_+. \tag{B.1}$$

Thus the implemented maximum positive all-node score gain minimizes exactly the resulting hard loss. All deletions are tested. Uniformly initializing that trial's weights changes its soft loss from hard loss by at most $n \log K$. Exact E/M fitting decreases soft loss; selecting the best trial together with the unchanged fit gives

$$\mathcal{F}_{t+1} \le (1 - 1/K)\mathcal{F}_t + Cn\sqrt{d} + n \log K, \qquad \mathcal{F}_{t+1} \le \mathcal{F}_t.$$

Initially $\mathcal{F}_0 \le Cnd$ for arbitrary simplex weights, since all profile divergences are $O(d)$ on the common box. Iterate the recurrence. If a complete round does not improve the fit, the same inequality gives $\mathcal{F}_t \le K(Cn\sqrt{d} + n \log K)$ directly. □

B.3 A deterministic E-step bootstrap

Lemma B.1 (Frobenius error and responsibility bootstrap)**.** *On the full-graph event of Lemma 3.6,*

$$\|\widetilde{A} - M_0\|_{\mathrm{F}}^2 \le Cnp, \qquad \|q - M_0\|_{\mathrm{F}}^2 \le Cnp.$$

If the fitted profiles have aligned ℓ_1 error $o(np)$ and $-\log \min_a \widehat{\pi}_a = o(np)$ (in particular, if the weights are bounded below), their next E-step satisfies

$$\frac{1}{n} \sum_{i,a} |R_{ia}^{\mathrm{new}} - Z_{ia}| \le \frac{C}{np}.$$

Proof. Best rank-K approximation in operator norm gives $\|A - \widetilde{A}\|_{\mathrm{op}} \le \|A - M_0\|_{\mathrm{op}} \le C\sqrt{np}$. The matrix $\widetilde{A} - M_0$ has rank at most $2K$, proving the Frobenius bound. True probabilities lie inside the clipping interval, so clipping cannot increase squared coordinate error.

The population gap $D(p_{a*} \| p_{b*})$ is at least $c_1 np$. On the common box, changing an observation or a parameter changes its score contrast by at most a constant times its ℓ_1 change. Thus a row with $\|q_i - p_{a*}\|_1 \le t_0 np$, for sufficiently small fixed t_0, has fitted correct-class margin at least $c_2 np$ for all large n. Its wrong responsibility is at most $(K-1)e^{-c_2 np}$. Every other row has squared ℓ_2 error at least $t_0^2 np^2$. The Frobenius bound limits their number to C/p, hence their fraction to $C/(np)$. □

For completeness, let $e = n^{-1} \sum_{l,a} |R_{la} - Z_{la}|$ and assume $e = o(1)$. Then $N_a = n_a + O(ne)$, and for $\bar{\mu}_a = n_a^{-1} \sum_{l : z_l = a} q_l$ the exact weighted mean identity gives

$$\|\mu_a - \bar{\mu}_a\|_1 \le Ce\, np, \qquad \|\bar{\mu}_a - p_{a*}\|_1 \le C\sqrt{np}.$$

Indeed, $\max_l \|q_l\|_1 \le Cnp$, while the second inequality follows by averaging the Frobenius estimate over a class of size at least $n/(\beta K)$. The updated weights equal $n_a/n + O(e)$ and are bounded below. Thus the first M-step has profile error $o(np)$. Therefore Lemma B.1 gives $R^{(1)}$ with error $C/(np)$. This refined radius remains invariant: an M-step from this radius has profile error at most $C\sqrt{np} + C = o(np)$ in ℓ_1. More quantitatively, choose a fixed small responsibility radius so that the resulting parameter error costs at most one quarter of the population margin. Every E-step from that fixed radius has error at most $C_*/(np)$, where C_* is independent of the previous error and of the iteration number. Since $C_*/(np)$ eventually lies inside the fixed radius, induction gives a common bound for all subsequent updates.

B.4 The selected endpoint already has the required accuracy

Proof of Lemma 3.5. Write $d = np$. Since $\mathcal{F} \geq \sum_i \min_a D(q_i \| \mu_a)$ and Bernoulli divergence on the box is at least $\|q_i - \mu_a\|_1^2/(Cd)$,

$$\sum_i \min_a \|q_i - \mu_a\|_1^2 \leq C\xi_n n d^2.$$

Also $\sum_i \|q_i - p_{z_i,*}\|_1^2 \leq n\|q - M_0\|_{\mathrm{F}}^2 \leq Cnd$. Choose $v_n = \xi_n^{1/4} + d^{-1/4} \to 0$. The fractions of rows farther than $v_n d$ from their true profile or from every fitted profile are at most $C/(v_n^2 d)$ and $C\xi_n/v_n^2$, respectively; both vanish. Each true class contains a row outside both sets. The K true profiles are separated in ℓ_1 norm by cd, so their nearby fitted profiles must be different. There are exactly K fitted profiles, giving the asserted common matching.

Fix a sufficiently small constant t_0. Except for Cn/d rows, the row's ℓ_1 distance to its true profile is at most $t_0 d$, by the Frobenius bound. For every such row in class a, matched-center consistency and box Lipschitz bounds imply

$$D(q_i \| \mu_b) \geq c_1 d \quad (b \neq a), \qquad D(q_i \| \mu_a) \leq c_1 d/4.$$

Each class has at least $n/(2\beta K)$ such rows for large n. Since each summand of $\mathcal{F}$ is nonnegative and on these rows is at least $-\log(\widehat{\pi}_a + e^{-c_1 d})$, we obtain

$$\widehat{\pi}_a + e^{-c_1 d} \geq e^{-2\beta K \xi_n d}, \qquad \widehat{\pi}_a \geq \tfrac{1}{2} e^{-2\beta K \xi_n d}$$

eventually. The latter holds whether or not $\xi_n d$ diverges, since $\xi_n \to 0$. Thus $-\log \widehat{\pi}_a \leq 2\beta K \xi_n d + \log 2 = o(d)$. Apply Lemma B.1 to the fit itself. □

C The spectral likelihood tail retains the oracle rate

The remaining argument compares error tails with the oracle LRT. It controls bounded changes of conditional cumulants and local tilted concentration. It does not round the spectral block sums to recover integer edge counts.

C.1 Fitted parameters and a bounded-moment score correction

Define the perfect-label fitted mean $\bar{\mu}_a = n_a^{-1} \sum_{l: z_l = a} q_l$. Let

$$\mathcal{G} = \max_{a,c} \left| \frac{\mathbf{1}_{C_a}^\top (A - M) \mathbf{1}_{C_c}}{\sqrt{n_a n_c}} \right|.$$

For each fixed t, $\mathbb{E} e^{t\mathcal{G}} \leq C_t$. Indeed, these are finitely many independent-edge sums with coefficient $O(n^{-1})$ and variance $O(p)$; the usual bounded-variable moment generating function estimate applies to both signs.

Lemma C.1 (Fitted-score comparison)**.** *On the full-graph good event, every M-step with aligned responsibility error at most $C/(np)$ satisfies, uniformly in i, a, b,*

$$\left| \ell_i(\mu_a) - \ell_i(\mu_b) + \log \frac{\widehat{\pi}_a}{\widehat{\pi}_b} - \left\{ \sum_j q_{ij} w_{z_j}^{ab} + \sum_j r_{z_j}^{ab} \right\} \right| \leq C(1 + \mathcal{G}). \tag{C.1}$$

Proof. The mean-fitting identity and the bound $\|q_l\|_1 \le Cnp$ show $\|\mu_a - \bar{\mu}_a\|_1 \le C$. Score derivatives in a parameter coordinate are $(q_{ij} - \mu_j)/[\mu_j(1-\mu_j)]$ and are bounded on the common box, so this replacement costs $O(1)$. Updated log-weight ratios are bounded as well.

Write $\delta_a = \bar{\mu}_a - p_{a*}$. Cauchy–Schwarz and the Frobenius bound give $\|\delta_a\|_2^2 \le Cp$. Also

$$\|q_i(I - \Pi)\|_2 \le C\sqrt{p}. \tag{C.2}$$

For $\widetilde{A}$, this follows from row incoherence, signal size $O(np)$, and projector error $O((np)^{-1/2})$. For q it follows because a common scalar 1-Lipschitz map contracts within-block variance: use the pairwise formula for variance in each block.

We next bound each block sum of δ_a. With $R = A - \widetilde{A}$, one has $R = (I - UU^\top)R(I - UU^\top)$ and $\|R\|_{\mathrm{op}} \le C\sqrt{np}$. Consequently

$$\|\Pi R \Pi\|_{\mathrm{op}} \le C\sqrt{np}\,\|\Pi - UU^\top\|_{\mathrm{op}}^2 \le C/\sqrt{np}.$$

Thus block-averaging $\widetilde{A}$ instead of A changes the relevant block sum by at most a constant. Raw block averaging contributes at most $C(\mathcal{G}+1)$, including the absent diagonal. Finally, if a true probability u is a distance at least $c'p$ from both clipping endpoints, then

$$|\operatorname{clip}(x, \varepsilon_n, 1-\varepsilon_n) - x| \le \frac{(x-u)^2}{c'p}.$$

Summing this inequality and averaging over class a gives $\|\bar{\mu}_a - n_a^{-1}\mathbf{1}_{C_a}^\top \widetilde{A}\|_1 \le C$. We have proved

$$\left|\sum_{j \in C_c} \delta_{aj}\right| \le C(1+\mathcal{G}). \tag{C.3}$$

Taylor-expand ℓ_i in its parameter around p_{a*}. Its second derivatives are bounded by C/p, so the quadratic remainder is at most $C\|\delta_a\|_2^2/p \le C$. Split the linear term into the true block span and its orthogonal complement. The block part is bounded by (C.3); the derivative coefficient after projection onto the true block span is block constant and bounded. The orthogonal part is bounded by $C\|q_i(I-\Pi)\|_2\|\delta_a\|_2/p \le C$ using (C.2). Apply this to both classes. □

C.2 An exact arrowhead reduction of empirical spectral block sums

Fix vertex i and condition on $B = A_{-i,-i}$. Its incident-edge vector $x = A_{-i,i}$ remains independent Bernoulli with block-constant probabilities. Let V, D be the retained signal eigenpairs of B, write $N = B - VDV^\top$, and let $\Pi_B = VV^\top$. Let Π_0 be the true block projector on the remaining vertices. On the vertex-deleted good event,

$$\|N\|_{\mathrm{op}} \le C\sqrt{np}, \quad \|V\|_{2\to\infty} \le C/\sqrt{n}, \quad \|\Pi_B - \Pi_0\|_{\mathrm{op}} \le C/\sqrt{np}.$$

All assertions below about x are deterministic on $\|x\|_1 \le D_0 np$, with fixed sufficiently large D_0.

Lemma C.2 (Arrowhead linearization with bounded remainder)**.** *For every fixed vector $w \in \mathbb{R}^n$ with $\|w\|_\infty \le W$,*

$$\left|\widetilde{A}_i w - x^\top \Pi_B w_{-i}\right| \le C_W. \tag{C.4}$$

The constant is uniform over all incident-edge vectors satisfying the degree bound.

Proof. Permute i to the first coordinate and write $A = \left(\begin{smallmatrix} 0 & x^\top \\ x & B \end{smallmatrix}\right)$. The difference from $\mathrm{diag}(0, B)$ has norm $\|x\|_2 \le \sqrt{D_0 np}$. Weyl's inequality consequently gives exactly K eigenvalues of magnitude $\asymp np$, with all remaining eigenvalues $O(\sqrt{np})$. This assertion uses good B and the degree bound alone. The retained eigenvectors can be written $u_r = (t_r, Vy_r + v_r)$ with $v_r \perp V$. The exact eigenvector equations give

$$v_r = t_r(\lambda_r - N)^{-1}x_Q, \qquad (\lambda_r - h_r)t_r = x_V^\top y_r, \quad h_r = x_Q^\top(\lambda_r - N)^{-1}x_Q,$$

where $x_V = V^\top x$, $x_Q = (I - \Pi_B)x$, and $|h_r| \le C$.

The padded basis $V_0 = (0, V^\top)^\top$ has spectral residual $AV_0 - V_0 D = (x_V^\top, 0)^\top$, of norm at most $Cnp/\sqrt{n}$. For a full-graph bulk eigenvector h and column $V_{0,r}$,

$$(\lambda_h - D_r)\langle h, V_{0,r}\rangle = \langle h, (AV_0 - V_0 D)_r\rangle.$$

The denominator has magnitude at least cnp, for either signal sign. Sum the squared identity over bulk eigenvectors and the fixed K columns to obtain $\|(I - UU^\top)V_0\|_{\mathrm{F}} \le C/\sqrt{n}$. Equality of the two subspace dimensions gives $\|UU^\top - V_0V_0^\top\|_{\mathrm{op}} \le C/\sqrt{n}$. In particular

$$\sum_r t_r^2 \le C/n, \qquad \left\|\sum_r y_r y_r^\top - I_K\right\|_{\mathrm{op}} \le C/n.$$

The latter uses squared principal angles, not the unsquared projector bound.

Put $w_V = V^\top w_{-i}$ and $w_Q = (I - \Pi_B)w_{-i}$. Expanding the retained spectral sum gives

$$\begin{aligned}\widetilde{A}_i w = x_V^\top \left(\sum_r y_r y_r^\top\right) w_V + \sum_r h_r t_r y_r^\top w_V \\ + \sum_r \lambda_r t_r^2 \{w_i + x_Q^\top(\lambda_r - N)^{-1} w_Q\}.\end{aligned}$$

The errors relative to $x_V^\top w_V$ are bounded respectively by

$$C(np/\sqrt{n})n^{-1}\sqrt{n} = Cp, \qquad Cn^{-1/2}\sqrt{n} = C, \qquad C(np/n)\{1 + \sqrt{np}\sqrt{n}/(np)\} \le C.$$

This proves (C.4) without assuming independence between the full-graph eigenvectors and x. □

Lemma C.3 (Projected block weights)**.** *For any true block-constant vector v on the remaining vertices with bounded entries, let $\beta = (\Pi_B - I)v$. Then*

$$\|\beta\|_\infty \le C, \quad \|\beta\|_2^2 \le C/p, \quad \left|\sum_{j \in C_c \setminus \{i\}} \beta_j\right| \le C/p.$$

Proof. The first follows from V's row bound and $\|v\|_2 = O(\sqrt{n})$, and the second from the projector bound. For the third, since $v \in \mathrm{col}(\Pi_0)$,

$$\Pi_0\beta = -\Pi_0(I - \Pi_B)\Pi_0 v, \qquad \|\Pi_0(I - \Pi_B)\Pi_0\|_{\mathrm{op}} \le \|\Pi_0 - \Pi_B\|_{\mathrm{op}}^2 \le C/(np).$$

Take its inner product with a block indicator. □

C.3 Clipping and the oracle tail prefactor

Choose a sufficiently small fixed $h > 0$ such that $2c_0 + h < \min(\kappa, \eta)/2$, and put

$$l = (2c_0 + h)p, \qquad u = 1 - (2c_0 + h)p.$$

For each full block C_c set $T_c = \widetilde{A}_i \mathbf{1}_{C_c}$ and $Y_c = x^\top \Pi_B \mathbf{1}_{C_c \setminus \{i\}}$.

Lemma C.4 (Deterministic buffered clipping envelopes)**.** *On the full-graph and vertex-deleted good events and the degree event,*

$$\sum_{j \in C_c} q_{ij} \geq \min(Y_c, n_c u) - C,$$
$$\sum_{j \in C_c} q_{ij} \leq \max(Y_c, n_c l) + C.$$

Proof. Write $\widetilde{A}_{ij} = T_c/n_c + v_j$ in block c. The v_j sum to zero and (C.2) for $\widetilde{A}$ gives $\sum v_j^2 \leq Cp$. If the block mean is at least hp below an upper clipping endpoint, its upper clipping loss is at most $\sum (v_j - hp)_+ \leq \sum v_j^2/(4hp) \leq C$. If it is above that buffered endpoint, monotonicity gives the same lower envelope with the block mean replaced by the buffered endpoint. The lower endpoint is treated symmetrically. Use $\varepsilon_n \leq 2c_0 p$ and then $|T_c - Y_c| \leq C$ from Lemma C.2. □

For a fixed pair a, b, abbreviate $w_c = w_c^{ab}$, $r_c = r_c^{ab}$, and define

$$H(x) = \sum_{w_c \geq 0} w_c \min(Y_c, n_c u) + \sum_{w_c < 0} w_c \max(Y_c, n_c l) + \sum_c n_c r_c.$$

The true-profile score based on q is at least $H(x) - C$. For each clipping pattern $S \subseteq [K]$, where S denotes the unclipped coordinates, its branch is

$$H_S(x) = x^\top \Pi_B v_S + \sum_{c \notin S} w_c k_c + \sum_c n_c r_c, \quad (v_S)_j = w_{z_j} \mathbf{1}\{z_j \in S\},$$

with $k_c = n_c u$ for $w_c \geq 0$ and $k_c = n_c l$ for $w_c < 0$. One has $H(x) = H_S(x)$ on the appropriate branch region; dropping that region gives an upper bound on each branch's error probability.

Lemma C.5 (Conditional cumulant and concentration bounds)**.** *Let $\alpha = \alpha^*_{i,ab}$. Uniformly over good B and every branch,*

$$\mathbb{E}_a[e^{-\alpha H_S(x)} \mid B] \leq C e^{-D^*_{i,ab}}. \tag{C.5}$$

If S contains a block with $|w_c| \geq \tau$, for a fixed $\tau > 0$, then, for every real t,

$$\mathbb{P}_a\{H_S(x) \leq t \mid B\} \leq \frac{C_\tau e^{-D^*_{i,ab} + \alpha t}}{\sqrt{np}}. \tag{C.6}$$

Proof. First replace $\Pi_B v_S$ by v_S. Write $m_c = n_c - \mathbf{1}\{z_i = c\}$. The ideal counts in different blocks are independent Binomial(m_c, P_{ac}). For sufficiently large n, the lower cutoff satisfies $n_c l \leq m_c P_{ac}$, including the deleted vertex's class, since $l < \kappa p/2$ and $n_c \geq n/(\beta K)$. The upper cutoff is above its mean as well. Thus each omitted coordinate satisfies $w_c k_c \geq w_c \mathbb{E}_a \sum_{j \in C_c \setminus \{i\}} x_j$. Jensen's inequality bounds its deterministic exponential factor by that block's full exponential moment. The ideal branch moment is at most $e^{-D^*_{i,ab} + O(p)}$: the branch and oracle intercepts differ by exactly $\sum_c (n_c - m_c) r_c = r_{z_i} = O(p)$.

For the empirical projection put $\beta = (\Pi_B - I)v_S$. Taylor-expand the log moment as a function of its coefficient vector. At the ideal block-constant coefficients its first derivative is a bounded multiple of a block-constant tilted probability, of order p. The block-sum bound in Lemma C.3 makes the linear change $O(1)$. Along the segment to the perturbed coefficients every second derivative is bounded by Cp, since all coefficients are bounded. The quadratic change is at most $Cp\|\beta\|_2^2 \le C$. This proves (C.5).

Under exponential tilting by $e^{-\alpha H_S}$ the coordinates of x are still independent Bernoulli, with probabilities comparable to p and complements bounded below. If a block in S has $|w_c| \ge \tau$, then $\|v_S\|_2^2 \ge c_\tau n$, while $\|\beta\|_2^2 \le C/p = o(n)$. The variance of $x^\top \Pi_B v_S$ under this tilted law is therefore between $c_\tau np$ and Cnp. Bounded coefficients give a total third absolute centered moment at most a constant times this variance. Berry–Esseen consequently bounds the probability of every interval of length one by $C_\tau/\sqrt{np}$.

Divide $(-\infty, t]$ into intervals $(t-r-1, t-r]$. Changing measure back gives

$$\mathbb{P}_a(H_S \le t \mid B) \le \mathbb{E}_a(e^{-\alpha H_S} \mid B)\frac{C_\tau e^{\alpha t}}{\sqrt{np}} \sum_{r \ge 0} e^{-\alpha r}.$$

The maximizing α is uniformly bounded away from zero. This proves (C.6); no nonlattice assumption or exact boundary density is used. □

Some branches can have no strongly weighted unclipped coordinate. They cannot cause an error on the graph event. In fact the oracle mean LLR is at least $c_1 np$. Omitted coordinates replace their means in the score-increasing direction. On $\|x\|_1 \le D_0 np$, $|Y_c| \le CD_0 np$. Therefore, if every unclipped weight has magnitude less than τ,

$$H_S(x) \ge c_1 np - K\tau(CD_0 + 1)np - O(1) \ge c_1 np/2$$

for a sufficiently small fixed τ. Thus all potentially erroneous branches have the concentration required by Lemma C.5.

C.4 Completing the refined risk bound

Let $\mathcal{G}_i$ denote the block fluctuation $\mathcal{G}$ computed on the vertex-deleted graph B, with the corresponding block sizes. On the good graph and degree event,

$$\mathcal{G} \le C(1 + \mathcal{G}_i), \qquad \mathcal{G}_i \le C\sqrt{np}, \qquad \mathbb{E}e^{t\mathcal{G}_i} \le C_t \quad (t \text{ fixed}).$$

The first bound holds because adding back one row changes a normalized block sum by at most $C \deg(i)/n + Cp \le C$.

After an M-step from the $C/(np)$ responsibility radius, Lemma C.1 shows that assigning vertex i from a to b implies

$$H(x) \le C(1 + \mathcal{G}_i).$$

First use the full-graph good event only to establish this deterministic containment of the error event. Then remove its indicator from the upper bound; it is not assumed independent of x. Conditional on good B, branches with only weak unclipped weights are impossible on the degree event for large n, since $\mathcal{G}_i = O(\sqrt{np})$. For every other branch apply (C.6), with threshold $C(1 + \mathcal{G}_i)$. Integrating over B and using its bounded exponential moment yields

$$\mathbb{P}_a\{\text{aligned assignment } i : a \to b, \text{ good events}\} \le C\frac{e^{-D^*_{i,ab}}}{\sqrt{np}}.$$

The union over $b \neq a$ and the at most 2^K branches costs a fixed constant. The graph, vertex-deleted graph, and degree failures are $C_H e^{-Hnp}$. Since $I_n \leq C_I np$, choose fixed $H > C_I + 1$; their total contribution is negligible relative to $e^{-I_n}/\sqrt{np}$. The conclusion is uniform over every fitted endpoint in the refined responsibility radius: the same graph event and the same fluctuation $\mathcal{G}_i$ control them all. Thus a data-dependent number of additional updates requires no union bound over iterations.

Proof of Theorem 3.7. An initial M-step from responsibilities of error $\gamma_n = o(1)$ has profile error $o(np)$ and weights bounded below, by the mean-fitting identity in Appendix B. Lemma B.1 makes the next E-step error at most $C/(np)$. The second M-step therefore satisfies Lemma C.1, and the conditional tail argument above gives (3.5). If the initial responsibilities already have error $C/(np)$, the first M-step itself meets that comparison.

Choose $H > C_I + 1$, where $I_n \leq C_I np$, and include the stipulated initialization failure. Since $np \to \infty$, all exceptional probabilities $C_H e^{-Hnp}$ are negligible relative to $e^{-I_n}/\sqrt{np}$. Summing over the fixed number of competing classes and averaging over vertices proves the expectation bound. The invariant radius in Lemma B.1, together with the comparison uniform over that radius, covers all further finite exact updates and finite data-dependent stopping rules. □

Proof of Theorem 3.1. Fix the same sufficiently large H. On G_H, all observed profiles and every convex-combination mean generated by the algorithm lie in the common box. Theorem 3.4 gives a deterministic uniform sequence $\xi_n \to 0$ such that the selected endpoint has $\mathcal{F} \leq \xi_n n^2 p$, regardless of the growing endpoint. Lemma 3.5 then shows that the algorithm's mandatory final E-step has error at most $C/(np)$. Its mandatory M-step satisfies the strong-initialization clause of Theorem 3.7. This proves (3.5), including initialization and estimated parameters, with no additional raw-edge access.

Finally $n\,\mathrm{Mis}$ is an integer, so $\mathbb{P}(\mathrm{Mis} > 0) \leq n\mathbb{E}\,\mathrm{Mis}$, giving the stated exact-recovery condition. The computational cost in Section 2 gives the polynomial-time assertion with polynomial fitting caps. □

C.5 Scope of the sparse-score comparison

For $p \leq 1/2$, the inequalities $|\log(1-u)+u| \leq Cu^2$ and $|\log(1-u) - \log(1-v)| \leq Cp$, for $u, v \leq p$, show that replacing the exact oracle Bernoulli contrast by the sparse contrast changes it by at most $C\{p\deg(i) + np^2\}$. A degree bound at any fixed confidence e^{-Hnp} makes this $O(np^2) = o(np)$ if $p \to 0$. This estimate does not establish the Zhou–Li affinity bound; that requires the bounded-cumulant analysis above. The dense example in the main text follows from the two Bernoulli large-deviation exponents at the stated thresholds.

D Implementation and reproducibility

D.1 Unregularized implementation

The source, complete numerical evidence and compiled manuscript are at

`https://github.com/zhixin0825/spectral-lrt`.

Algorithm 1 is implemented by `SpectralLikelihood.decode()` in

`experiments/20261010_general_density/spectral_likelihood.py`.

It accepts an $n \times K$ eigenvector array and a length-K array of the corresponding signed eigenvalues. This unregularized decoder receives neither the adjacency matrix, true labels, true block probabilities nor node degrees. Retaining the top-$|\lambda|$ eigenpairs is its preceding spectral step. The general decoder below additionally retains the observed degree vector.

The default `score_family='bernoulli'` evaluates (2.13). The floor coefficient defaults to $c_0 = 10^{-4}$; its theoretical guarantee requires the explicit admissibility condition in Section 2, not merely positive block probabilities. The spectral scale is $\max\{1, \max_a |\lambda_a|\}$, and the reconstructed entries are clipped on both sides. The original adjacency is not regularized. The optional `score_family='sparse'` is a score control on precisely the same q, and is not the default general-density algorithm.

Growing uses at most 100 exact M-steps per stage. A fit stops after at least one M-step if hard labels are unchanged and its absolute soft-loss change is at most $10^{-9}n$. Replacement uses at most $\lceil \log n \rceil$ rounds, tries every component deletion against the same incumbent, initializes each trial's weights uniformly, and uses one M-step per trial by default. A full round with no improvement stops the search. This rule is justified by the contraction certificate in Appendix B. Calling `decode()` always executes the final E-step and one profile/weight M-step before returning labels.

The implementation uses the unconstrained simplex weight update. Zero-mass components retain their previous profile. Log-sum-exp arithmetic evaluates responsibilities and the soft divergence loss F without exponentiating very negative raw log-likelihoods. Theoretical divergences are nonnegative; the code clips negative floating-point residue at zero. Spectral block sums are not rounded to integers. Weighted means, including the initial global mean, are clipped back to the same feasible interval solely to correct roundoff in convex combinations. The theorem is an exact-arithmetic result; recorded numerical checks assess implementation accuracy to floating-point precision.

The user may increase either fitting budget. Explicitly requesting fewer replacement rounds is an experimental option and does not automatically retain the stated general-K initialization guarantee. The proven default budget is $\lceil \log n \rceil$, with valid early stopping after a full unsuccessful round.

D.2 Degree-plus-spectrum implementation

Algorithm 2 is implemented by `DegreeSpectralLikelihood.decode()` in

`experiments/20261018_degree_spectral/degree_spectral_likelihood.py`.

It receives only the degree vector and the signed eigenpairs of WAW. Its reconstruction uses a Moore–Penrose inverse of $[\mathbf{1}, WU]$, allowing redundant retained directions. The data matrix is neither symmetrized nor clipped, preserving its exact degree identity. Candidate and fitted probability profiles are clipped to the parameter interval; the constrained M-step is the clipped weighted mean. Scores and the soft objective are evaluated directly by the complete Bernoulli criterion and log-sum-exp arithmetic, since signed reconstructed data need not admit a nonnegative Bernoulli KL representation.

The reproducible check is

`experiments/20261018_degree_spectral/verify_degree_spectral.py`.

It compares the compressed reconstruction against the full-graph identities, including redundant feature directions and signed reconstructed entries, and checks exact-EM objective monotonicity. Saved checks are in the same directory. The graph is used to generate and verify test fixtures; the decoder itself receives only (D, U, Λ). These are algebraic and implementation checks, separate from an unproved unconditional sparse-rate claim.

D.3 Current finite-graph and population evidence

The complete 60-graph design is specified by

`experiments/20261010_general_density/benchmark_results/protocol.json`.

It fixes all graph seeds, model families, density settings, method parameters, and hashes of the code used at fitting time. Each graph directory retains the original A, P, true labels for evaluation, exact retained (U, Λ), all fitted profiles and weights, every candidate's hard loss, and full EM objective traces. Ground truth is not used for fitting, selecting replacements, or stopping. The aggregate CSV contains all 360 graph–method combinations.

The archived fitting engine is retained as `engine_at_fitting.py`. A subsequent boundary fix clips the rounded initial mean when a graph has all reconstructed entries at the floor. The maintenance record checks that this expression is bitwise unchanged on every archived graph, so the saved benchmark is unaffected. The original fitting hashes remain in the frozen protocol; maintenance provenance is recorded separately.

The five-community population calculation and the exact rational interval certificate are in

`experiments/20261010_general_density/population_audit/`.

They record the positive-definiteness certificate, strict all-node-gain comparisons, the limiting one-update-per-stage trajectory, and replacement trial results. The distinction between a population limiting path, a finite-graph theorem, and fully iterated growing is maintained in the report. The manifest `SHA256SUMS` identifies the saved numerical records.

D.4 Historical experiments and entry points

The earlier 240-graph comparison is preserved at `experiments/20261009_residual_likelihood_seeding/`. The `residual_seed.py` module retains several sparse-score methods, using

$$\ell_i^{\mathrm{sparse}}(\widehat{p}) = \sum_j q_{ij} \log \widehat{p}_j - \sum_j \widehat{p}_j.$$

Its current default `method='global_gain_certified'` calls the separate sparse threshold-certificate implementation `Work.iterative_certified_growing()`. It uses an adaptive lower-only floor, a uniform-weight refit of the current centers as well as K replacement trials, acceptance threshold $(\sum_{ij} q_{ij})/\sqrt{\log n}$, and a cap of $\lceil (\log n)^2 \rceil$ accepted rounds. Its general-K proof for the sparse range $p \to 0$ with $p = \Omega(\log n/n)$ and its 24-graph implementation checks are retained as a companion version. They preceded the complete Bernoulli density extension and direct soft-loss contraction presented here, which allow the full range (2.3).

In that module, `method='global_gain'` is the growing-only branch, `method='global_gain_restarts'` selects independent residual starts, and `method='repair'` is the historical single-pass repair. None identifies the current full-Bernoulli Algorithm 1, whose entry point is specified above. The actual 240-graph runs used the earlier lower floor $10^{-4} \log n/n$ and their recorded protocol, not the adaptive-floor default now present in the companion module. Their profiles were not constrained to lie below one. The preserved benchmark runner explicitly selects `floor_mode='historical'` for these original arms; their earlier source is also accessible in repository history.

The archived 240-graph fits use an exact weight M-step with floor $f = 10^{-8}$. For a fixed $0 \le f < 1/K$, it solves

$$\max_{\pi_a \ge f,\, \sum_a \pi_a = 1} \sum_a N_a \log \pi_a, \qquad \pi_a = \max\{f, N_a/\lambda\}, \quad \sum_a \pi_a = 1.$$

An active-set solver determines λ; naive clipping and renormalization is not this optimizer. The current main theorem and implementation use $f = 0$.

Historical fits stop only when labels are unchanged, relative profile and absolute weight changes are at most 10^{-3}, and the objective increment is at most $10^{-6}(1+|\mathcal{L}-G|)$, where $G = \sum_{ij} \log \Gamma(q_{ij}+1)$ is an observation-only tolerance constant. The cap is 300 updates. Historical top-subset and peeling controls, their subset sizes and stopping rules are defined by the archived code. They are not steps of Algorithm 1.

D.5 Build and validation

The active manuscript contains six main sections, statistical and deterministic proof appendices, implementation details, and references. PDFLaTeX, BibTeX and two further PDFLaTeX passes resolve the bibliography and cross references. Build provenance records the source revision and source/PDF checksums. The manuscript is rendered for a page-by-page layout check. These syntax, reference and layout checks are separate from the likelihood-tail proof and numerical evidence described above. The current theorem preserves the oracle rate up to constants; no exact integer-count or risk-ratio-one claim is used in the implementation or interpretation.

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